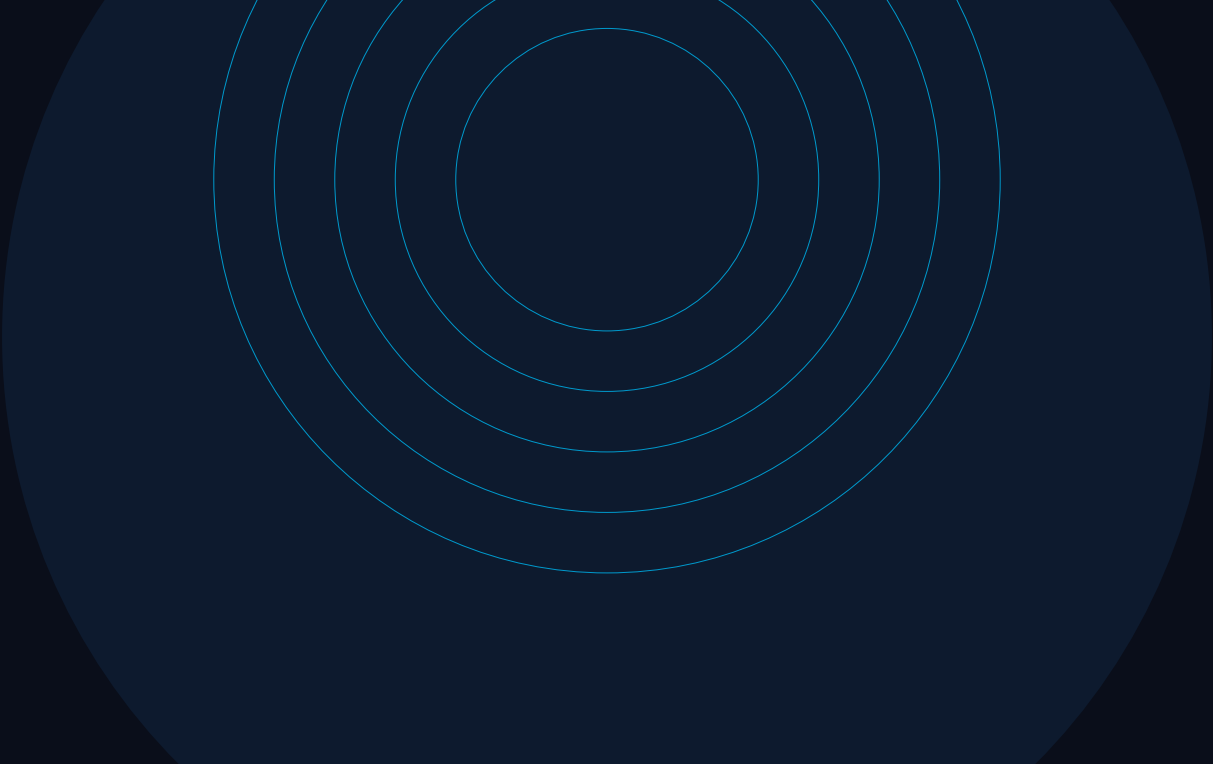




RTL-SDR V3 Console

A Comprehensive User Guide

[Installation](#) - [Features](#) - [Decoders](#) - [Troubleshooting](#) - [Reference](#)

091.500000 MHz WFM 2.40 Msps

Version 1.0

[jwfrancois](#) / [Z.ai](#)

github.com/jwfrancois/rtl-sdr-v3-console

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Chapter 1: Introduction & Quick Start

The **RTL-SDR V3 Console** is a high-definition web application for software-defined radio. Built with Next.js, TypeScript, and the Web Audio API, it transforms the popular RTL-SDR V3 USB dongle into a complete radio workstation with spectrum analysis, waterfall display, multiple demodulators, and decoders for aircraft, weather satellites, pagers, GPS, and more. The app runs entirely in a browser, with an optional WebSocket bridge that connects to your physical hardware through the standard `rtl_tcp` daemon.

What sets this console apart from other SDR software is its dual-mode architecture: it ships with a fully-functional **simulated SDR engine** that synthesizes realistic spectrum data and audio for 29 stations across the FM, airband, marine, NOAA, ham, and shortwave bands. This means you can explore every feature, learn how to tune, and experiment with decoders without any hardware at all. When you are ready to receive real signals, a one-click toggle switches the app to your physical RTL-SDR V3 dongle via the bridge script, and the same UI now displays live RF from your antenna.

Two Operating Modes

The console supports two backends that you can switch between at any time using the Hardware Source panel in the top-left of the screen:

Simulated Mode (default)

The simulated SDR engine runs entirely in your browser. It generates believable spectrum data with station peaks, noise floor, ionospheric fading on HF frequencies, and per-bin Gaussian noise. The Web Audio synthesizer produces station-appropriate audio: layered chord music for FM broadcast stations, filtered noise for voice-like stations, keyed tones for CW/morse, alternating FSK for data stations, and so on. This mode is perfect for learning the UI, demonstrating the app, or testing features without any hardware.

Real Hardware Mode

When you switch to real mode, the app connects to a WebSocket bridge running on your PC. The bridge wraps `rtl_tcp` and forwards raw IQ samples to the browser, where they are processed by the same FFT, demodulators, and decoders. Every frequency change, gain adjustment, or demodulator switch is sent back to the bridge and applied to the hardware in real time. You hear real broadcast audio, see real aircraft on the radar plot, and decode real RDS data from local FM stations.

Quick Start: 5 Minutes to Live Audio

If you just want to hear something right now, follow these steps. No hardware required.

Path A: Simulated Mode (no hardware)

1. Open the app in your browser (the preview URL is shown in the development environment).
2. The app starts in simulated mode by default, tuned to 91.5 MHz in WFM mode.
3. Click the **AUDIO ON** button in the transport bar (next to START/STOP).
4. You will hear synthesized jazz music from the Jazz Horizon station.
5. Click different frequencies in the spectrum or use the digit tuner (left panel) to switch stations. Each station type plays different audio.

Tip: Try clicking on the spectrum at 127.2 MHz (AM mode) to hear simulated air traffic control, or 162.4 MHz (NFM) for NOAA weather radio. Each station has a different audio character.

Path B: Real Hardware Mode

Connecting real hardware takes about 10 minutes if you have Node.js installed. The full setup is covered in Chapter 5; here is the short version:

```
# Terminal 1 - talk to the dongle
sudo rtl_tcp -s 2400000

# Terminal 2 - run the bridge (download from the app)
cd ~/sdr-bridge
npm install
npm start

# In the web app:
# 1. Click 'Real RTL-SDR' in the Hardware Source panel (top-left)
# 2. Wait for status to flip to 'LIVE HW' (about 1 second)
# 3. Click AUDIO ON to hear real broadcast audio
```

If the bridge shows `rtl_tcp handshake OK`, you are connected. The spectrum will now show real RF from your antenna instead of simulated data.

Browser Requirements

The console uses modern Web APIs that require a recent browser:

Feature	API	Minimum Browser
WebSocket client	WebSocket API	All modern browsers
Real-time FFT canvas	Canvas 2D + requestAnimationFrame	Chrome 60+, Firefox 55+, Safari 11+
Audio synthesis	Web Audio API	Chrome 60+, Firefox 55+, Safari 14+
High-DPI rendering	devicePixelRatio	All modern browsers
Pointer events	Pointer Events API	Chrome 55+, Firefox 59+, Safari 13+

Mobile browsers work but the layout is condensed. The app is optimized for desktop use with a screen width of at least 1280 pixels. For the best experience, use a recent version of Chrome, Firefox, or Edge on a desktop or laptop.

Warning: Browsers block insecure WebSocket (`ws://`) connections from secure HTTPS pages, with one exception: `ws://localhost` and `ws://127.0.0.1` are always allowed. If you are running the cloud preview and want to access the dongle on a different PC, you need to start the bridge with the `--tls` flag for `wss://` support. See Chapter 5 for details.

Chapter 2: Installing & Running the App

This chapter covers everything you need to run the RTL-SDR V3 Console on your own machine. The app is a standard Next.js 16 project, so it runs on any operating system with Node.js 18 or later. You can run it locally for development, or deploy it to any static hosting provider that supports Next.js (Vercel, Netlify, Cloudflare Pages).

Prerequisites

Before you begin, make sure you have the following installed:

Tool	Version	Purpose
Node.js	18.0 or later	JavaScript runtime for the dev server
npm or bun	npm 9+ / bun 1.0+	Package manager
Git	any recent version	Clone the repository
RTL-SDR drivers	rtl-sdr 0.6+	Only needed for real hardware mode

If you plan to use real-hardware mode, you also need the RTL-SDR package installed on your system, which provides the `rtl_tcp` daemon. Installation instructions for macOS, Linux, and Windows are in Chapter 5.

Cloning the Repository

The full source code is available on GitHub. Clone it to your local machine:

```
git clone https://github.com/jwfrancois/rtl-sdr-v3-console.git
cd rtl-sdr-v3-console
```

The repository contains the complete application: the Next.js app in `src/`, the standalone bridge script in `download/sdr-bridge/`, and documentation in the README. The bridge is also served from `public/download/sdr-bridge/` so it can be downloaded directly from the running app.

Installing Dependencies

Install the JavaScript dependencies using your preferred package manager:

```
# Using npm (default)
npm install

# Or using bun (faster, recommended)
bun install
```

This will install Next.js 16, React 19, TypeScript 5, Tailwind CSS 4, the shadcn/ui component library, Zustand for state management, and all other dependencies. The install takes about 30 seconds with npm or 10 seconds with bun.

Starting the Dev Server

Run the development server:

```
npm run dev # or: bun run dev

# The app starts on http://localhost:3000
# Open this URL in your browser to see the console
```

The first time you load the page, Next.js will compile the project (this takes a few seconds). Subsequent loads are instant thanks to Turbopack. The app starts in simulated mode by default, so you should immediately see a spectrum with simulated stations and be able to click around to explore.

Project Structure Overview

Understanding the project layout helps if you want to extend the app or contribute:

```
rtl-sdr-v3-console/
src/
app/ Next.js App Router
globals.css Dark theme, glassmorphic panels
layout.tsx Root layout + font setup
page.tsx Main dashboard page
components/
ui/ shadcn/ui components
sdr/ SDR-specific panels
spectrum-analyzer.tsx
waterfall-display.tsx
frequency-tuner.tsx
... (20+ panel components)
lib/
sdr-engine.ts Simulated SDR engine
sdr-store.ts Zustand state store
sdr-audio.ts Web Audio synth + real audio
real-sdr/
types.ts Protocol types
dsp.ts FFT, biquads, decimation
demodulators.ts FM/AM/SSB/CW
rds.ts RDS decoder
adsb.ts ADS-B Mode S decoder
apt.ts NOAA APT decoder
meteor.ts Meteor M2 LRPT
goes-hrit.ts GOES HRIT
inmarsat-stdc.ts Inmarsat STD-C
gps-l1.ts GPS L1 C/A
pocsag.ts POCsag pagers
acars.ts ACARS aircraft messaging
hd-radio.ts HD Radio SIS
notch-filter.ts Notch filter
download/sdr-bridge/ Standalone bridge script
prisma/ Database (unused by default)
public/ Static assets
```

Building for Production

To create a production build (faster page loads, smaller bundle):

```
npm run build
npm run start # serves the built app on port 3000
```

The production build is optimized and minified. You can deploy the built app to any Next.js-compatible host. Vercel is the easiest - just push to GitHub and connect your repo on vercel.com, and Vercel will deploy automatically on every commit.

Chapter 3: The User Interface Tour

The console uses a three-column dashboard layout designed to fit a 1280-pixel-wide screen comfortably. On narrower screens the columns stack vertically. Each panel addresses a specific function, and most panels become important only after you switch to real-hardware mode and tune to the right frequency band. This chapter walks through every panel and what it does.

Top Header (Status Bar)

Across the top of the page, the status header shows live system information: the active frequency in large monospace text, the active station name (if you are tuned to a known frequency), the band name (FM Broadcast, Airband, Marine VHF, etc.), the current demod mode, the sample rate in Msps, the tuner gain, the PPM correction, the SNR estimate, and the system clock with uptime. On the right are status badges that show SDR ON/OFF, AUDIO ON/MUTED, and REC/IDLE states.

Left Column: Controls

Hardware Source Panel

At the very top of the left column, this panel lets you switch between the simulated engine and your real RTL-SDR hardware. The two-button toggle selects the backend; below it, the bridge URL field lets you point the app at a bridge running on another machine. When connected to real hardware, the panel shows live stats: device name (typically RTL-SDR V3 R820T2), uptime, current hardware frequency, sample rate, gain, and any buffer overruns. A green LIVE HW badge indicates a healthy connection.

Frequency Tuner

Below the connection panel is the VFO (Variable Frequency Oscillator) tuner. The frequency is shown as large monospace digits grouped by GHz, MHz, kHz, and Hz. Each digit is individually clickable: left-click increments that digit, right-click decrements it. Below the digits are Up/Down buttons that tune by the selected step size (10 Hz to 1 MHz). Six quick-band preset buttons let you jump to common frequencies with the correct demodulator mode and bandwidth pre-configured.

Demodulator Controls

Seven demodulator mode buttons let you select WFM, NFM, AM, USB, LSB, CW, or RAW. Below the mode buttons are bandwidth presets appropriate to the selected mode. For example, WFM offers 180 kHz through 240 kHz (broadcast FM channel widths); NFM offers 12.5 kHz through 30 kHz (VHF/UHF voice channel widths); AM offers 6 kHz through 16 kHz (suitable for airband and shortwave broadcast). The selected bandwidth is shown as amber dashed lines on the spectrum display, indicating the receiver filter window.

RF & Audio Panel

This panel controls the radio frequency chain: tuner gain (0-50 dB or Auto), sample rate (eight presets from 240 ksps to 3.2 Msps), squelch threshold (0-100 percent), audio volume, PPM correction (-100 to +100), and AGC speed (slow, medium, fast). The AGC checkbox toggles between manual gain control (where you set the dB value) and automatic gain control (where the dongle's AGC circuit adjusts gain to maintain a constant output level).

Notch Filter Panel

The notch filter lets you suppress strong interfering signals in the IQ stream before demodulation. This is most useful when a strong local FM station is swamping your dongle's front end and preventing you from hearing weaker signals. The Auto-detect toggle continuously scans the spectrum for bins 25 dB above the noise floor and automatically adds notches at those frequencies. You can also add manual notches by entering a frequency offset in Hz and clicking the + button. The active notch list shows each notch with its frequency offset, Q factor, and an auto/manual badge.

Scanner Panel

The scanner automates signal discovery. Three modes are available: Peak scan finds the strongest signal in the current spectrum view and tunes to it every 800 ms. Sweep scan moves through a band in 70 percent sample-rate steps, collecting every signal above -60 dBFS into a sorted list. Squelch scan steps through a band in channel-sized increments, stops for 3 seconds on each signal above the squelch threshold, then continues. Nine band presets cover FM broadcast, airband, 2m ham, NOAA weather, marine VHF, 70cm ham, GMRS, shortwave, and CB radio. Found signals are listed with frequency and signal strength; click any entry to tune back to it.

Center Column: Visualization

Transport Bar

The transport bar sits at the top of the center column. It contains the START/STOP button (controls whether the SDR is producing IQ data), the AUDIO ON/OFF button (starts the Web Audio context and routes demodulated audio to your speakers), the master volume slider, the REC button (toggles IQ recording on the bridge), and a squelch indicator that shows SQL OPEN (signal above threshold) or SQL CLOSED (signal below threshold). When real audio is being received and played, an additional activity meter appears showing the peak audio amplitude and the audio frame rate.

Spectrum Analyzer

The HD spectrum analyzer shows the magnitude spectrum of the current IQ stream in real time. The X axis is frequency (labeled in MHz, kHz, or Hz depending on the span); the Y axis is dBFS, ranging from -100 to 0. A glowing cyan line shows the instantaneous spectrum, a yellow peak-hold line shows recent peak values, and a gradient fill under the curve gives a visual sense of signal strength. The amber dashed lines indicate the demodulator filter bandwidth, and the yellow triangle at the top marks the tuned center frequency. You can click anywhere on the spectrum to tune to that frequency instantly.

Waterfall Display

Below the spectrum, the waterfall display scrolls downward over time, showing the history of the spectrum as a color-coded image. The Viridis colormap runs from dark purple (low signal) through blue, teal, green, yellow, and finally bright yellow (strong signal). The waterfall's vertical axis is time, with the most recent data at the top. This display is invaluable for spotting transient signals: an aircraft transponder burst appears as a brief streak, while a continuous FM station appears as a solid vertical line. The filter bandwidth and center frequency markers carry over from the spectrum above.

Signal Meter (S-Meter)

The S-meter shows the current signal strength on a traditional S0 through S9+60 dB scale, with a green-yellow-orange-red gradient bar. The squelch threshold is shown as a red dashed marker. This mimics the S-meter on a traditional communications receiver: S9 corresponds to approximately -73 dBm, and each S-unit is 6 dB. The meter is useful for comparing antenna positions or for confirming that a signal is above the squelch threshold.

Audio Oscilloscope

The audio oscilloscope shows the frequency-domain representation of the demodulated audio output. When audio is enabled and the SDR is producing real audio frames, you see the spectrum of what you are hearing - typically a peak at the dominant audio frequency. This is useful for verifying that the demodulator is producing audio and for spotting interference or distortion in the audio chain.

Decoder Panels (auto-show)

Below the signal meter, additional decoder panels appear automatically when you tune to the right frequency band. The ADS-B tracker appears at 1090 MHz with a radar-style polar plot of decoded aircraft. The APT weather satellite panel appears at 137-138 MHz (when not on a NOAA APT frequency), showing the live image as it is decoded. GOES HRIT, Inmarsat STD-C, Meteor M2, and GPS L1 panels appear at their respective frequencies. See Chapters 8 through 17 for decoder-specific details.

Right Column: Stations & Recording

Active Station Card

This card shows information about the station currently tuned (in simulated mode) or the strongest signal under the cursor (in real mode). It displays the station name, band, modulation, bandwidth, and description, along with a live signal strength percentage, the exact tuned frequency, and the current demod mode. An animated radio wave backdrop adds visual interest.

IQ Recording Panel

When connected to real hardware, this panel lets you record raw IQ data to disk on the bridge. The big RECORD/STOP button toggles recording; live stats show duration, file size, and write rate in MB/s. Below the active recording stats, a list of saved recordings appears with size, timestamp, and a download link. Files are saved as raw unsigned 8-bit IQ (.raw), compatible with SDR#, GQRX, and GNU Radio.

Audio WAV Recorder

Below the IQ recording panel, the audio WAV recorder captures the demodulated audio as a 16-bit PCM WAV file directly in the browser. This is useful for capturing a quick clip of what you are hearing without dealing with raw IQ data. The RECORD AUDIO button starts capture; clips are listed with duration, file size, download link, and delete button.

Decoded Messages Panel

This panel shows decoded POCSAG pager messages and ACARS aircraft messaging. The tab toggle switches between Pagers and ACARS views. Pager messages are tagged NUM (numeric), ALN (alphanumeric), or TON (tone-only) with the pager address and message text. ACARS messages show the flight ID, aircraft registration, message label, message number, and text body. The panel auto-switches to the appropriate tab when you tune to a pager band (929-932 MHz) or ACARS band (131-132 MHz).

Memory Bank (Bookmarks)

The memory bank has three tabs: Stations, Bookmarks, and History. The Stations tab lists 29 pre-configured real-world frequencies grouped by band (FM Broadcast, Airband, Marine VHF, Weather, Ham Radio, Shortwave, etc.) with a search field to filter by name, band, or frequency. The Bookmarks tab lets you save the current frequency/demod/bandwidth combination for later recall. The History tab lists up to 24 recently tuned frequencies. When connected to real hardware, cloud icons in the panel header let you push your bookmarks to the bridge or pull them on another device, enabling cross-device bookmark sync.

Floating Elements

RDS / HD Radio Overlays

When tuned to a broadcast FM station in WFM mode, two overlay cards appear on top of the spectrum panel. The RDS overlay (top-right) shows the station's PS name (the 8-character name you see on a car radio display), PI code, PTY (program type), Radio Text, and other RDS data. The HD Radio overlay (top-left) shows similar data from HD Radio SIS: call letters, slogan, FCC facility ID, ALFN (GPS-locked time), and audio service code. Both overlays require real hardware and a strong FM signal.

Keyboard Shortcuts Button

A small keyboard icon floats in the bottom-right corner of the viewport. Click it (or press the ? key) to open the keyboard shortcuts help overlay. See Chapter 22 for the complete shortcut reference.

Fullscreen Spectrum Mode

Click the maximize icon in the top-right corner of the spectrum panel to enter fullscreen mode. The spectrum grows to 300 pixels tall and the waterfall to 400 pixels, both spanning the full viewport width. The RDS overlay continues to show in the top-right. Press ESC or click the EXIT button to leave. This mode is useful for monitoring a band for an extended period or for demonstrations.

Chapter 4: Tuning & Demodulation

This chapter covers the fundamental operations of any SDR: tuning to a frequency and selecting the right demodulator. The RTL-SDR V3 Console provides multiple ways to tune, each suited to different use cases, and seven demodulator modes that cover the vast majority of signals you will encounter between 24 kHz and 1.75 GHz.

Ways to Tune

Digit-by-Digit Tuning

The frequency tuner in the left column shows the current frequency as 12 monospace digits grouped as GGG.MMM.kkk.HHH (gigahertz, megahertz, kilohertz, hertz). Each digit is independently clickable: hover over a digit to reveal small up/down arrows, or simply click the digit itself to increment it by one (right-click decrements). This is the most precise way to tune and is ideal for entering an exact frequency from a list.

Step Up/Down Buttons

Below the digits are UP and DOWN buttons that tune by the selected step size. The step size is shown in a dropdown between the buttons, with eight options: 10 Hz, 100 Hz, 1 kHz, 5 kHz, 12.5 kHz, 25 kHz, 100 kHz, and 1 MHz. The 12.5 kHz and 25 kHz steps correspond to common VHF/UHF voice channel spacing, while 100 kHz is useful for scanning the FM broadcast band. Use the keyboard arrow keys (up/down) to step quickly without reaching for the mouse.

Click on Spectrum or Waterfall

The fastest way to explore a band is to click directly on the spectrum or waterfall display. The X position of your click is mapped to a frequency, and the app tunes there instantly. You can also drag (hold the mouse button down and move) to scrub across the band. As you hover, the current cursor frequency is shown in the panel header. This is the recommended way to find unknown signals.

Quick Band Presets

Below the step buttons are six quick-band preset buttons: FM (96.5 MHz WFM), Airband (127.2 MHz AM), Marine (156.8 MHz NFM), NOAA (162.4 MHz NFM), 20m Ham (14.2 MHz USB), and WWV (10 MHz AM). Each preset tunes to a representative frequency in that band and configures the demodulator mode and bandwidth appropriately. Use these as starting points for exploring each band.

Memory Bank Stations List

The Stations tab in the right column lists 29 real-world frequencies grouped by band. Each entry shows the station name, description, modulation, and exact frequency. Clicking any entry tunes to it with the correct demod and bandwidth. Use the search field to filter by name, band, or frequency. This is the fastest way to jump to a specific known signal.

Demodulator Modes

The RTL-SDR V3 Console supports seven demodulator modes, each suited to a specific class of signal. Selecting the wrong mode will produce garbled or silent audio. The following table summarizes when to use each mode.

Mode	Typical Use	Bandwidth	Audio Cutoff
WFM	Broadcast FM (87.5-108 MHz)	180-240 kHz	15 kHz
NFM	VHF/UHF voice (marine, ham, NOAA)	12.5-30 kHz	3 kHz
AM	Airband, shortwave broadcast	6-16 kHz	4 kHz
USB	HF ham radio voice (above 10 MHz)	2.4-3.5 kHz	3 kHz
LSB	HF ham radio voice (below 10 MHz)	2.4-3.5 kHz	3 kHz
CW	Morse code	300-1000 Hz	800 Hz tone
RAW	Magnitude output (for decoders)	240 kHz - 2 MHz	varies

WFM (Wide FM)

WFM demodulates broadcast FM signals with 75 kHz deviation and a 15 kHz audio cutoff. The demodulator differentiates the phase of each IQ sample to recover the instantaneous frequency, then normalizes by the expected deviation. A 75 microsecond de-emphasis filter (2122 Hz cutoff) compensates for the pre-emphasis applied at the transmitter, restoring flat frequency response. The audio is then decimated to 48 kHz for playback. Use WFM for any signal in the 87.5-108 MHz broadcast band.

NFM (Narrow FM)

NFM handles narrowband FM signals with 5 kHz deviation and 3 kHz audio cutoff. The demodulator is identical to WFM but uses a smaller deviation value and a tighter audio low-pass filter. Use NFM for VHF/UHF voice communications: marine VHF (156-162 MHz), NOAA weather radio (162.4-162.55 MHz), 2m and 70cm ham radio, GMRS, and business band radios.

AM (Amplitude Modulation)

AM demodulates by computing the magnitude of each complex IQ sample, then subtracting a slowly-tracked DC average to remove the carrier. The result is the audio envelope. A low-pass filter at 4 kHz removes high-frequency noise and the carrier residual. Use AM for the aircraft band (118-137 MHz), shortwave broadcast (2-30 MHz), and time signal stations like WWV (2.5, 5, 10, 15, 20 MHz).

USB / LSB (Single Sideband)

Single-sideband demodulation uses the phasing method: the IQ signal is mixed with a cosine and sine wave at a small offset frequency (500 Hz BFO), and the result is low-pass filtered to remove the unwanted sideband. USB keeps the upper sideband (frequencies above the carrier); LSB keeps the lower sideband. Use USB for ham voice above 10 MHz (20m, 15m, 10m bands); use LSB for ham voice below 10 MHz (80m, 40m bands). The 500 Hz BFO shifts the audio into the audible range so you hear a natural-sounding voice instead of DC.

CW (Continuous Wave / Morse)

CW demodulation extracts the magnitude envelope of the signal, then mixes it with a 700 Hz BFO (beat frequency oscillator) to produce an audible tone whenever the carrier is present. A bandpass filter from 500 Hz to 900 Hz removes noise outside the tone frequency. Use CW when receiving Morse code signals in the ham bands (typically around 14.0-14.070 MHz for 20m CW, but you can find CW activity in all ham bands).

RAW

RAW mode does not demodulate - it outputs the magnitude of the IQ stream directly. This is useful for decoders that need the raw baseband signal, such as ADS-B (aircraft transponders at 1090 MHz use a 1 Mbps PPM modulation that requires magnitude-domain processing). RAW output is rarely useful for human listening but is essential for the ADS-B and other specialized decoders.

Bandwidth Selection

Each demodulator mode has a set of appropriate bandwidth presets. The bandwidth determines the receiver filter width - signals within this window are demodulated; signals outside are attenuated. A wider bandwidth captures more of the signal but also more noise; a narrower bandwidth rejects adjacent-channel interference but may truncate the audio. As a rule of thumb: match the bandwidth to the signal. Broadcast FM uses 180 kHz; narrow FM voice uses 12.5 or 25 kHz; AM broadcast uses 10 kHz; SSB voice uses 2.4 or 3 kHz.

The selected bandwidth is visualized on the spectrum as amber dashed lines, showing exactly which portion of the spectrum is being demodulated. This is helpful when adjusting bandwidth to match an unknown signal - widen until the audio sounds clear, then narrow until just before the audio quality degrades.

Gain Control

The tuner gain controls how much the RTL-SDR's R820T2 tuner amplifies the incoming RF signal before digitization. There are 29 gain steps from 0 to 49.6 dB. Higher gain lets you hear weaker signals but also amplifies noise; lower gain reduces noise but may miss weak signals. The right setting depends on your antenna, the band, and local interference levels.

AGC (Automatic Gain Control)

When AGC is enabled (checkbox in the RF & Audio panel), the tuner automatically adjusts gain to maintain a constant output level. This is convenient for general listening but can cause gain pumping on strong signals (the gain drops when a strong signal appears, then rises when it disappears, modulating the background noise). For weak-signal work or when comparing signals, disable AGC and set gain manually.

Manual Gain Tips

Start with AGC on to find the band. Once you have a signal of interest, disable AGC and increase gain until the noise floor rises noticeably on the spectrum (typically around 30-40 dB). Then back off 5 dB to leave headroom for signal peaks. If you see clipping (the spectrum hits 0 dBFS frequently), reduce gain. If the noise floor is below -90 dBFS, you can safely increase gain.

PPM Correction

The RTL-SDR V3's crystal oscillator has a typical accuracy of ± 50 PPM (parts per million), meaning the actual sample rate and tuned frequency can be off by up to ± 50 Hz per MHz of frequency. At 100 MHz, that is ± 5 kHz - enough to noticeably shift the pitch of demodulated audio. The PPM correction slider lets you compensate for this error.

To calibrate PPM: tune to a known exact frequency (e.g. a local FM station's licensed frequency from the FCC database). Listen to the demodulated audio and adjust PPM until it sounds natural. For broadcast FM, the easiest reference is a station that carries RDS: when the RDS PI code decodes successfully, your PPM is close enough. Once you find your dongle's PPM offset, it stays constant for all frequencies (it is a property of the crystal, not the tuned band).

Squelch

The squelch threshold mutes the audio output when the signal level drops below a configurable level. This prevents the constant hiss of background noise during periods of no signal. The squelch slider in the RF & Audio panel sets the threshold from 0 to 100 percent of full scale. The S-meter shows the squelch threshold as a red dashed marker, so you can see at a glance whether the current signal is above or below the threshold.

Set squelch just above the noise floor: open the squelch fully (0 percent) and observe the S-meter reading during silence. Then set squelch to just above that level. The transport bar's SQL indicator shows OPEN (signal present) or CLOSED (signal absent). For scanners, squelch is used to detect signals: the squelch scan mode stops on each frequency where the signal exceeds the threshold.

Chapter 5: Connecting Real Hardware

This chapter walks through the complete process of connecting your physical RTL-SDR V3 dongle to the web console. The setup involves three components: the `rtl_tcp` daemon (talks to the USB dongle), the bridge script (wraps `rtl_tcp` and exposes it over WebSocket), and the web app (connects to the bridge). Once all three are running, you can receive real RF signals from your antenna.

Step 1: Install `rtl_tcp`

The `rtl_tcp` daemon ships with the open-source RTL-SDR package. Installation instructions vary by operating system.

macOS (Homebrew)

```
brew install librtlsdr
```

This installs the `librtlsdr` library and the `rtl_tcp` command-line tool. The dongle is immediately accessible to non-root users - no `udev` rules needed on macOS.

Linux (Debian/Ubuntu)

```
sudo apt install rtl-sdr
```

After installation, you need to set up `udev` rules so non-root users can access the USB device. See the "USB Permissions" section below.

Linux (Arch)

```
sudo pacman -S rtl-sdr
```

Windows

Download prebuilt binaries from osmocom.org or rtl-sdr.com. The zip file contains `rtl_tcp.exe` and the necessary DLLs. Extract to a folder of your choice and add it to your `PATH` for convenience. Windows does not require `udev` rules.

Step 2: USB Permissions (Linux only)

On Linux, the RTL-SDR dongle appears as a USB device with VID:PID `0bda:2838`. By default, only root can access raw USB devices. You need to install a `udev` rule that grants read/write permission to your user.

```
# Write udev rules covering RTL-SDR V3 + V4 + common clones
sudo tee /etc/udev/rules.d/20.rtlsdr.rules > /dev/null <<'EOF'
SUBSYSTEMS=="usb", ATTRS{idVendor}=="0bda", ATTRS{idProduct}=="2838", MODE=="0666",
GROUP="plugdev"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="0bda", ATTRS{idProduct}=="2832", MODE=="0666",
GROUP="plugdev"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="1d50", ATTRS{idProduct}=="6089", MODE=="0666",
GROUP="plugdev"
EOF

sudo udevadm control --reload-rules
sudo udevadm trigger
```

```
# Add yourself to the plugdev group, then LOG OUT and back in
sudo usermod -aG plugdev $USER
```

After running these commands, you MUST log out and log back in (or reboot) for the group change to take effect. Then unplug and replug the dongle. If you see "usb_open error -3" when running rtl_tcp, the rules have not taken effect - see the troubleshooting section at the end of this chapter.

Tip: As a quick fallback that bypasses udev entirely, you can run rtl_tcp as root: `sudo rtl_tcp -s 2400000`. The bridge itself does not need root - only rtl_tcp does.

Step 3: Start rtl_tcp

Plug in your RTL-SDR V3 and open a terminal. Run rtl_tcp with a default sample rate of 2.4 Msps (you can change this later from the web app):

```
rtl_tcp -s 2400000
```

You should see output similar to:

```
Found 1 device(s):
0: Realtek, RTL2838UHIDIR, SN: 00000001

Using device 0: Generic RTL2832U OEM
Found Rafael Micro R820T tuner
[R82XX] PLL not locked!
Listening on TCP port 1234...
```

If you see "usb_open error -3" or "Failed to open rtl_sdr device #0", your user does not have permission to access the USB device. Revisit Step 2 above, or use the sudo fallback: `sudo rtl_tcp -s 2400000`.

Step 4: Download and Run the Bridge

The bridge script wraps rtl_tcp and exposes the IQ stream over WebSocket so the browser can connect. The bridge is a single Node.js file with one dependency (the ws WebSocket library).

Download the bridge

Open the web app and look for the Hardware Source panel in the top-left corner. Click the "Real RTL-SDR" button. The panel expands to show a download link for the bridge script. Click "Download bridge script" to save bridge.mjs to your Downloads folder.

Alternatively, download directly from the running app:

```
mkdir -p ~/sdr-bridge
cd ~/sdr-bridge
# If the app is running locally:
curl -O http://localhost:3000/download/sdr-bridge/bridge.mjs
curl -O http://localhost:3000/download/sdr-bridge/package.json
curl -O http://localhost:3000/download/sdr-bridge/README.md
```

Install dependencies and start

```
cd ~/sdr-bridge
npm install # installs the ws WebSocket library
npm start # starts the bridge on port 8080
```

When the bridge starts, you should see:

```
[bridge] WebSocket server listening on ws://0.0.0.0:8080
[bridge] HTTP download server listening on http://0.0.0.0:8081/recordings
[bridge] connecting to rtl_tcp at 127.0.0.1:1234...
[bridge] connected to rtl_tcp
[bridge] rtl_tcp handshake OK (tuner: RTL-SDR V3 (R820T2))
```

If you see the handshake message, the bridge is successfully connected to your dongle. The bridge is now waiting for the web app to connect.

Step 5: Connect from the Web App

In the web app's Hardware Source panel, ensure "Real RTL-SDR" is selected and the bridge URL is `ws://localhost:8080` (the default). The status indicator should flip from OFFLINE to CONNECTING to LIVE HW within about one second. The bridge log will show a new "client connected" message.

Click the AUDIO ON button in the transport bar. You should hear demodulated audio from the current frequency. If you are still tuned to 91.5 MHz in WFM mode (the default), you will hear whatever FM station is broadcasting at that frequency in your area - or static if no station is there. Click around the spectrum to find a strong peak, then enable AUDIO to hear it.

Portable Access with TLS (`wss://`)

By default the bridge speaks plain `ws://` (insecure WebSocket). Browsers allow `ws://localhost` from any page, but if you want to access the dongle from another device - your phone, another PC, or the cloud preview - you need `wss://` (TLS).

Option A: Self-signed certificate (LAN access)

Start the bridge with the `--tls` flag; it auto-generates a self-signed certificate:

```
node bridge.mjs --tls
# Bridge now listens on wss://0.0.0.0:8443
# Certificate saved to ./certs/cert.pem and ./certs/key.pem
```

Your browser will warn about the self-signed certificate the first time. Visit `https://your-pc-ip:8443/` once in your browser and click "Advanced -> Proceed" to trust it. After that, the WebSocket URL `wss://your-pc-ip:8443` works from any device on your LAN.

Option B: Your own certificate

If you have a Let's Encrypt certificate or a corporate CA, pass the paths directly:

```
node bridge.mjs --tls --cert /path/fullchain.pem --key /path/privkey.pem
```

Option C: ngrok tunnel (anywhere access)

If you want to access the dongle from anywhere in the world, use ngrok to tunnel the bridge:

```
# Terminal 1 - your bridge
node bridge.mjs

# Terminal 2 - expose via ngrok
ngrok http 8080
# Outputs: https://abcd-1234.ngrok-free.app
```

Then in the web app, set the bridge URL to `wss://abcd-1234.ngrok-free.app` (replacing with your actual ngrok URL). This works from any browser anywhere in the world and tunnels through your firewall automatically.

Troubleshooting Connection Issues

usb_open error -3

This is a Linux USB permission error. Your user cannot access the raw USB device. Apply the udev rules from Step 2 above, log out, and log back in. If that fails, use the sudo fallback: `sudo rtl_tcp -s 2400000`. The bridge itself does not need root - only `rtl_tcp` does.

Bridge shows "rtl_tcp connection closed, reconnecting"

The dongle may have been unplugged, or another SDR app is holding it. Close any other SDR software (SDR#, GQRX, CubicSDR), then restart `rtl_tcp`. The bridge will automatically reconnect within 2 seconds once `rtl_tcp` is back.

Web app shows OFFLINE even with bridge running

Check the bridge URL in the Hardware Source panel. If the bridge is on a different PC, you need to enter that PC's IP address (e.g. `ws://192.168.1.50:8080`). If you are accessing the web app via HTTPS, you must use `wss://` and start the bridge with `--tls`. Browsers block `ws://` from HTTPS pages except to localhost.

No audio even with signal present

Verify the audio activity meter in the transport bar shows non-zero values. If it shows 0 percent, the demodulator is producing silence - check that you are on the correct demod mode (WFM for broadcast FM, AM for airband, etc.) and that the signal is centered in the filter window (amber dashed lines on the spectrum). If the meter shows activity but you hear nothing, check your browser's audio output settings and system volume.

Audio sounds clipped or choppy

The browser's WebSocket throughput tops out around 2-3 MB/s on a clean LAN. At 2.4 Msps, that is 4.8 MB/s of IQ data - too much for some connections. Try a lower sample rate: 1.024 Msps or 0.24 Msps. The bridge will report overruns in the connection panel if it is dropping samples.

Tuner shows "RTL-SDR (tuner 83886080)" instead of "R820T"

This was a bug in an earlier version of the bridge that read the `rtl_tcp` handshake with the wrong byte order. The number 83886080 is 5 in big-endian (which is the R820T tuner ID). Update to the latest bridge version (re-download `bridge.mjs` from the Connection panel). The fix reads the handshake correctly and displays "RTL-SDR V3 (R820T2)" as expected.

Chapter 6: The Bridge - Protocol & Internals

The bridge is a small Node.js script that connects to `rtl_tcp` on one side and exposes the IQ stream over WebSocket on the other. It also runs an HTTP server for serving recorded IQ files and syncing bookmarks. This chapter documents the wire protocol so you can write your own client, extend the bridge, or integrate it into other software.

Command Line Options

Flag	Default	Description
<code>--rtl-host</code>	127.0.0.1	Host where <code>rtl_tcp</code> is running
<code>--rtl-port</code>	1234	Port <code>rtl_tcp</code> is listening on
<code>--ws-host</code>	0.0.0.0	Bind address for the WebSocket server
<code>--ws-port</code>	8080	Port for the WebSocket server
<code>--tls</code>	(off)	Enable TLS (<code>wss://</code>). Auto-generates a self-signed cert.
<code>--cert</code>	(auto)	Path to TLS certificate (with <code>--tls</code>)
<code>--key</code>	(auto)	Path to TLS private key (with <code>--tls</code>)
<code>--http-port</code>	8081	Port for the HTTP server (recordings, presets)

WebSocket Protocol

The bridge speaks a simple JSON + binary protocol over a single WebSocket connection.

Client to Server (JSON text frames)

The browser sends JSON commands to control the SDR:

```
{"type": "set_frequency", "hz": 91500000}
{"type": "set_sample_rate", "hz": 2400000}
{"type": "set_gain", "db": 30}
{"type": "set_gain", "db": "auto"}
{"type": "set_ppm", "ppm": 0}
{"type": "start"}
{"type": "stop"}
{"type": "status"}
{"type": "start_recording"}
{"type": "stop_recording"}
```

Server to Client (JSON text frames)

The bridge periodically broadcasts status messages (every 500 ms) and sends recording-complete notifications:

```
{"type": "status", "payload": {
  "connected": true, "deviceName": "RTL-SDR V3 (R820T2)",
  "frequency": 91500000, "sampleRate": 2400000,
  "gainDb": 30, "ppm": 0, "gains": [],
  "overruns": 0, "uptime": 42.3,
  "recording": null
}}
```

```
}}
```

Server to Client (binary frames)

IQ data is sent as binary frames with a 16-byte little-endian header followed by raw IQ bytes:

Offset	Type	Field
0	uint32 LE	Sample rate in Hz
4	uint32 LE	Frequency - low 32 bits (Hz)
8	uint32 LE	Frequency - high 32 bits (always 0 for RTL-SDR)
12	uint32 LE	Timestamp (ms, truncated to 32 bits)
16+	bytes	Interleaved unsigned 8-bit I/Q pairs (I,Q,I,Q,...)

Each IQ sample is two bytes: I (in-phase) and Q (quadrature), both unsigned 8-bit centered on 127. To convert to signed float, subtract 128 and divide by 128. The frames are sent in 32 KB chunks (16K complex samples) every 50 ms, giving a steady data rate of about 640 KB/s at 2.4 Msps.

HTTP Server Endpoints

The bridge also runs an HTTP server on port 8081 (ws_port + 1) for file operations:

Endpoint	Method	Description
/health	GET	Returns JSON {ok:true, ws:..., port:...}
/recordings	GET	Lists saved IQ recordings (JSON array)
/recordings/{name}	GET	Downloads a specific recording file
/presets	GET	Returns saved bookmarks JSON
/presets	PUT	Saves bookmarks JSON to presets.json

RTL-SDR Command Bytes

The bridge forwards commands to rtl_tcp using its native binary protocol. Each command is 5 bytes: 1 byte command code + 4 bytes big-endian value. The command codes used by the bridge are:

Code	Constant	Description
1	SET_FREQUENCY	Tune to a frequency in Hz
2	SET_SAMPLE_RATE	Set the sample rate in Hz
3	SET_GAIN_MODE	0>manual, 1=auto (AGC)
4	SET_GAIN	Manual gain in tenths of a dB (e.g. 300 = 30.0 dB)
5	SET_FREQ_CORRECTION	PPM correction (signed)

The full rtl_tcp command set is documented at the [osmocom.org](http://osmocom.org/wiki) wiki. The bridge implements only the subset needed by the web app; additional commands (like SET_TUNER_GAIN_BY_INDEX or GET_TUNER_GAINS) can be added by extending the switch statement in the bridge's message

handler.

Recording Format

When you click RECORD in the IQ Recording panel, the bridge starts writing raw IQ bytes to a file in the `recordings/` directory next to the bridge script. The file format is header-less: just raw interleaved unsigned 8-bit I/Q pairs, exactly as received from `rtl_tcp`. Files are named with the format: `iq-{ISO timestamp}-{frequency}MHz-{sampleRate}ksps.raw`.

These files are compatible with SDR#, GQRX, GNU Radio, and most other SDR software. To replay a recording in SDR#: select File -> Play, choose "Raw IQ File", set the sample format to "8-bit unsigned", and enter the original sample rate. In GNU Radio, use a File Source block with item type "complex" and vector length 1, then multiply by 1/128 and subtract 1 to convert to float.

Extending the Bridge

The bridge is a single 280-line JavaScript file with no build step. To add a new command (for example, to control an external hardware filter), edit the `wss.on('message', ...)` handler in `bridge.mjs`, add a new case to the switch statement, and send the appropriate `rtl_tcp` command. On the client side, extend the `SdrCommand` type in `src/lib/real-sdr/types.ts` and call `source.configure()` with your new command.

Chapter 7: Decoders Overview

The console ships with 11 signal decoders that automatically activate when you tune to the appropriate frequency band. Each decoder runs on the IQ stream in real time, extracting structured data (text, images, aircraft positions, etc.) and displaying it in a dedicated UI panel. This chapter provides a quick-reference table of all decoders; the following chapters cover each one in detail.

Decoder Reference Table

Decoder	Frequency	Signal Type	Antenna	Difficulty
RDS	87.5-108 MHz	FM broadcast data	Stock whip	Easy
HD Radio SIS	87.5-108 MHz	FM broadcast (digital)	Stock whip	Medium
ADS-B	1090 MHz	Aircraft transponder	1090 MHz antenna	Easy
NOAA APT	137.1/137.5/137.9 MHz	Weather satellite	V-dipole or QFH	Medium
Meteor M2 LRPT	137.1/137.9 MHz	Weather satellite (digital)	V-dipole or QFH	Medium
GOES HRIT	1685.7/1694.1 MHz	Geostationary weather	Helical or dish + LNA	Hard
POCSAG	929-932 MHz, 138-174 MHz	Pagers	Stock whip or 900 MHz Yagi	Easy
ACARS	131.55 MHz	Aircraft messaging	Stock whip	Easy
Inmarsat STD-C	1537-1545 MHz	Satellite messaging	Helical or patch + LNA	Hard
GPS L1 C/A	1575.42 MHz	GPS positioning	Active GPS antenna	Medium
Notch filter	Any band	Interference removal	N/A	Easy

How Decoders Auto-Activate

Each decoder checks the current tuned frequency on every IQ block. If the frequency falls within the decoder's band of interest AND at least one UI component has subscribed to that decoder's updates, the decoder runs. This subscription-gated design means decoders only consume CPU when their results are actually being displayed - if you scroll past the ADS-B panel, the ADS-B decoder stops running to save battery and CPU.

What Each Decoder Produces

Decoders produce one of four output types:

Text overlays

RDS and HD Radio show floating overlay cards on the spectrum panel with station name, program type, and other metadata. These update roughly once per second.

Live image displays

APT and Meteor M2 show the decoded image growing line-by-line as the satellite passes overhead. A full pass takes 10-15 minutes and produces a strip image of the Earth below the satellite's path.

Tabular data lists

ADS-B shows a list of tracked aircraft with callsign, altitude, speed, position, and a radar-style polar plot. POCSAG and ACARS show a scrolling list of decoded messages with timestamps. GPS shows a per-PRN satellite list with signal strength and tracking status.

File assembly

GOES HRIT assembles received files from CCSDS CADU frames, showing a progress bar as each file is received and a list of completed files when done. Files are categorized by type (Image, Text, JPEG-LS, EMWIN, etc.).

Realistic Expectations

Not every decoder will produce results on every signal. Reception depends on your antenna, location, time of day, and atmospheric conditions. The ADS-B decoder will show planes whenever your antenna can see them (typically within 100-200 nm). The APT decoder only produces images during the 12-minute window when a NOAA satellite is overhead. The GPS decoder requires an active GPS antenna (with built-in LNA) - the stock RTL-SDR whip cannot pick up GPS. See each decoder's chapter for specific antenna recommendations and troubleshooting tips.

Chapter 8: RDS (Radio Data System)

RDS is a 1187.5 bps digital subcarrier injected at 57 kHz on broadcast FM signals. It carries station identification, program type, radio text, and other data that you typically see on a car radio display. The console's RDS decoder runs automatically whenever you tune to a station in the FM broadcast band (87.5-108 MHz) in WFM mode, and displays the decoded data as a floating overlay card on the spectrum panel.

What RDS Carries

The RDS protocol defines several data groups. The most common are:

Code	Name	Description
PI	Program Identification	4-hex code uniquely identifying the station
PS	Program Service	8-character station name (e.g. BBC R4)
PTY	Program Type	0-31 code (News, Rock, Jazz, etc.)
RT	Radio Text	Up to 64 chars of scrolling text (song title, etc.)
CT	Clock Time	UTC time and date from the station
TA	Traffic Announcement	Flag indicating traffic news is being broadcast
TP	Traffic Program	Flag indicating station carries traffic info
MS	Music/Speech	Flag indicating current content type
AF	Alternative Frequencies	List of other frequencies carrying the same station

How to Receive RDS

RDS reception is straightforward - it works with the same antenna and settings you use for broadcast FM:

1. Switch to real-hardware mode (see Chapter 5) and connect to your dongle.
2. Tune to a strong local FM station in WFM mode. The spectrum should show a clear peak with good signal-to-noise ratio.
3. The RDS overlay appears automatically in the top-right corner of the spectrum panel after a few seconds. If the station carries RDS, you will see the PS name (large text), PI code, PTY, and any Radio Text the station is broadcasting.

Tip: If the RDS overlay shows "Waiting for RDS data" for more than 10 seconds, the signal is too weak or the station does not carry RDS. Try a different station or adjust your antenna for better reception.

How the Decoder Works

The RDS decoder performs the following steps for each IQ block received:

- 1. Mix down by 57 kHz.** The RDS subcarrier sits at exactly 57 kHz (the third harmonic of the 19 kHz stereo pilot). We multiply the IQ stream by a 57 kHz cosine to bring the subcarrier to baseband.

2. Bandpass filter ± 2.4 kHz. The RDS signal occupies a narrow band around 57 kHz. A biquad bandpass filter removes everything outside this range.

3. DBPSK demodulation. RDS uses Differential Binary Phase-Shift Keying: each bit is encoded as a 180-degree phase flip (or not) relative to the previous bit. We multiply each sample by the previous sample to recover the bit stream.

4. Bit synchronization. The decoder looks for the 16-bit sync pattern (alternating 1s and 0s) at the start of each RDS group.

5. CRC-10 validation. Each 26-bit RDS block includes a 10-bit CRC (using the polynomial $x^{10} + x^8 + x^7 + x^5 + x^4 + x^3 + 1$). Blocks with invalid CRC are discarded.

6. Group assembly. A complete RDS group consists of 4 blocks (104 bits). The decoder assembles blocks with consecutive syndrome positions (0, 1, 2, 3) into a complete group.

7. Parameter decoding. Group type 0 contains the PS name (4 groups needed to assemble all 8 characters). Group type 2 contains Radio Text (16 groups for the full 64-character message).

Interpreting the Display

The RDS overlay shows the following fields when available:

PS Name (large text)

The 8-character Program Service name - what you see on a car radio display. This is the most prominent field. It updates character-by-character as each group 0 is received, so the name may appear partially garbled for the first few seconds.

PI Code

The 4-hex Program Identification code uniquely identifies the station. The first character is the country code; the remaining 3 characters are the station reference. You can look up PI codes in the FCC FM database.

PTY (Program Type)

The 5-bit Program Type code (0-31). Common values: 1=News, 5=Education, 10=Pop Music, 11=Rock Music, 12=Easy Listening, 14=Other Music, 17=Weather, 23=Travel. The full table is in the IEC 62106 standard.

Radio Text

Up to 64 characters of free-form text. Typically shows the current song title and artist, or a station slogan. The text scrolls character-by-character as new groups arrive, so it may take 10-30 seconds for the full message to appear.

Group Count

The total number of valid RDS groups decoded since the decoder was reset. A healthy RDS stream produces about 10-20 groups per second. If the count is not increasing, the signal is too weak or the station is not broadcasting RDS.

Troubleshooting

No RDS overlay appears

Verify you are in WFM mode and tuned to a frequency in the 87.5-108 MHz band. The overlay only shows when in real-hardware mode with HW connected. Check the connection panel shows LIVE HW.

Overlay shows "Waiting for RDS data" indefinitely

The signal is too weak or the station does not carry RDS. Try a stronger station (look for a tall peak on the spectrum). Some community and college stations do not broadcast RDS.

PI code is correct but PS shows garbage

The PS name is sent 2 characters at a time, so it takes 4 group 0 messages to complete. If reception is marginal, you may see a mix of old and new characters. Wait a few seconds for the full message to arrive. If the problem persists, adjust your antenna for better signal.

Chapter 9: HD Radio (NRSC-5)

HD Radio is the digital broadcasting standard used in North America for FM and AM stations. It transmits digital audio and data alongside the analog signal, providing CD-quality audio, multiple subchannels, and text data. The console's HD Radio decoder focuses on the SIS (Station Information Service) sub-channel, which carries station identification, call letters, and time synchronization.

Why SIS Only?

Full HD Radio audio decoding would require implementing the HDC (High-Definition Coding) audio codec, which is proprietary and has no open-source decoder. Even the open-source nrsc5 project, which decodes HD Radio, had to reverse-engineer parts of the codec. Instead, we decode the SIS sub-channel, which uses standard BPSK modulation and provides useful station information without the audio codec headache. You hear the analog FM audio; the SIS overlay shows the digital station ID.

HD Radio Signal Structure

An HD Radio FM signal consists of the analog FM carrier plus several digital sidebands:

Subband	Offset	Bandwidth	Content
P1	±10.7 kHz	±8.0 kHz	Primary audio (main program)
P2	±10.7 kHz	±8.0 kHz	Secondary audio (subchannel)
PIDS	±29.0 kHz	±5.0 kHz	Station Information Service (SIS)
P3	±20.4 kHz	±4.0 kHz	Additional subchannels

All sidebands fit within the standard 200 kHz FM channel allocation. Your RTL-SDR's WFM filter (180 kHz bandwidth) captures all of them. The SIS decoder focuses on the PIDS subband at ±29 kHz.

What SIS Provides

The SIS decoder extracts the following data from the PIDS sub-channel:

Field	Description	Example
Country	3-bit country code (US, CA, MX, etc.)	US
Facility ID	19-bit FCC facility identifier	12345
Call Letters	4-character station call sign	WNYC
Slogan	Up to 56 chars of station name	WNYC-FM New York
ALFN	Absolute Frame Number (GPS-locked time)	2024-01-15 12:34:56
Audio Service	Service descriptor (MP1, MP2, MP3)	MP1

How to Receive HD Radio

HD Radio reception uses the same antenna and setup as standard FM broadcast:

1. Connect your RTL-SDR V3 and switch to real-hardware mode.
2. Tune to a strong local FM station in WFM mode. In the US, most stations broadcast HD Radio alongside their analog signal.
3. The HD Radio overlay appears automatically in the top-left corner of the spectrum panel (alongside the RDS overlay in the top-right). After a few seconds, you should see the call letters and ALFN time stamp appear.
4. Click AUDIO ON to hear the analog FM audio (digital audio decoding is not supported).

Tip: HD Radio is primarily a North American standard. In Europe, the equivalent is DAB/DAB+, which uses different frequencies (174-240 MHz) and a different protocol. This decoder does not handle DAB.

How the Decoder Works

The SIS decoder performs the following steps:

- 1. Mix down by 29 kHz.** The PIDS subband sits at ± 29 kHz from the center frequency. We multiply the IQ stream by a 29 kHz cosine to bring the PIDS to baseband.
- 2. Bandpass filter ± 5 kHz.** The PIDS bandwidth is ± 5 kHz. A biquad low-pass removes everything outside this range.
- 3. DBPSK demodulation at 352.7 bps.** SIS uses differential BPSK at 352.7 bits per second (1 bit per OFDM symbol, 1 ms per symbol).
- 4. Sync word detection.** The SIS frame begins with a 32-bit sync word (0x7C95E5E8). The decoder searches for this pattern (and its bit-inverse, to handle the differential decoding ambiguity).
- 5. CRC-16 validation.** Each SIS frame includes a 16-bit CRC (CCITT polynomial 0x1021). Frames with invalid CRC are discarded.
- 6. Message parsing.** The 96-bit SIS payload contains one of four message types: SIS params (country + facility ID), station ID (call letters), station name (slogan), or ALFN + audio service descriptor.

What to Expect

RDS and HD Radio SIS provide similar information, but HD Radio SIS is more comprehensive. If both decoders are active, you will see both overlays on the spectrum panel. The HD Radio overlay is typically more reliable than RDS (HD Radio has stronger FEC) and provides additional data like the ALFN time stamp and facility ID that RDS does not carry.

Warning: Some stations broadcast HD Radio but do not populate all SIS fields. You may see the call letters but no slogan, or vice versa. This is normal - the station simply isn't broadcasting that particular data.

Chapter 10: ADS-B Aircraft Tracking

ADS-B (Automatic Dependent Surveillance-Broadcast) is the system by which aircraft broadcast their position, altitude, velocity, and identification twice per second on 1090 MHz. The console decodes these Mode S Extended Squitter messages and displays the aircraft on a radar-style polar plot, with a list showing callsign, altitude, speed, and position. This is one of the most rewarding decoders to use because aircraft are almost always within range.

ADS-B Signal Structure

Aircraft broadcast ADS-B on 1090 MHz using 1 Mbps PPM (Pulse Position Modulation) with the following frame structure:

Field	Bits	Description
Preamble	8	Synchronization pattern (2 pulses, 1 us apart)
DF (Downlink Format)	5	17=ADS-B, 18=Non-transponder, 11=Short
CA (Capability)	3	Aircraft capability flags
ICAO Address	24	Unique 24-bit aircraft address
ME (Message)	56	Type-specific data (position, velocity, etc.)
PI (Parity/Interrogator)	24	Error correction + interrogator ID

The ME field contains different data depending on the 5-bit type code at its start: type codes 1-4 carry the aircraft callsign (8 characters), type codes 9-18 carry airborne position (using Compact Position Reporting - CPR), and type code 19 carries airborne velocity. CPR is a clever scheme that encodes global latitude and longitude in just 35 bits each by alternating between two encodings (odd and even) that must be received in pairs.

How to Receive ADS-B

ADS-B reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. Tune to 1090 MHz. The demodulator mode does not matter (RAW is recommended - the ADS-B decoder works on raw IQ samples, not demodulated audio).
3. The ADS-B panel appears automatically in the center column below the signal meter. If you don't see it, verify you are tuned to 1090 MHz $\pm$ 10 MHz.
4. Set sample rate to 2.4 Msps (default) for best results. The 1 Mbps signal needs at least 2 Msps to capture cleanly.
5. Set gain to 30-40 dB (manual, AGC off) or use AGC. Higher gain can help with weak signals from distant aircraft.

Antenna Recommendations

The stock RTL-SDR whip antenna works for ADS-B if you are within 50-100 nm of an airport or major flight path. For better range, a dedicated 1090 MHz antenna is recommended. Options include:

Antenna Type	Range	Cost	Notes
Stock whip (vertical)	50-100 nm	Included	OK for beginners
1/4 wave ground plane	100-150 nm	\$10-15 DIY	Easy to build
Collinear (8-element)	150-250 nm	\$25-40	Best DIY option
Commercial 1090 MHz antenna	200-300 nm	\$30-60	FlightAware Pro Stick Plus
Pre-amplified antenna	250+ nm	\$50-100	Includes LNA for weak signals

Tip: Place your antenna near a window with a clear view of the sky. ADS-B signals are line-of-sight, so obstructions (buildings, trees, terrain) limit range. The aircraft must be above your local horizon to be received.

Interpreting the Radar Plot

The ADS-B panel includes a polar radar plot showing all tracked aircraft relative to your receiver. The receiver is at the center (cyan dot). Range rings are drawn at 50, 100, 150, and 200 nautical miles. Aircraft appear as small triangles oriented in the direction of travel, labeled with their callsign. A slow rotating sweep beam adds a traditional radar aesthetic.

The center of the plot is computed as the centroid of all decoded positions, which approximates your receiver's location (since ADS-B range is typically 100-200 nm, the centroid is close to the receiver). Aircraft appear when their position is decoded and disappear after 60 seconds without a new message.

Aircraft List

Below the radar plot, the aircraft list shows each tracked aircraft with:

Field	Description	Example
ICAO	24-bit hex aircraft address	A1B2C3
Callsign	Flight identifier (from type 1-4)	UAL123
Altitude	Barometric altitude in feet	37000 ft
Position	Latitude, longitude (from CPR)	40.7128, -74.0060
Speed	Ground speed in knots	450 kt
Track	Heading in degrees (0-359)	270
Vertical Rate	Climb/descent in ft/min	+500 (climbing)

The list is sorted by most recent message received. Aircraft that have not been heard from in 60 seconds are removed automatically. Click any aircraft in the list to highlight it on the radar plot.

Verifying with Flightradar24

To verify your ADS-B decoding, compare the aircraft you see with the live map at flightradar24.com. The callsigns, altitudes, and positions should match. Discrepancies usually indicate a weak signal causing position decode errors, or the occasional false positive from the preamble detector (these are filtered out by the 60-second timeout).

Performance Notes

The ADS-B decoder processes about 100-500 messages per second from a typical antenna in a busy area. Each message is 120 microseconds long, so the duty cycle is low. The decoder uses a sliding-window preamble detector that runs in real time on a modern CPU. If you see high CPU usage, try lowering the sample rate to 1.024 Msps - this reduces the workload without significantly affecting ADS-B decode quality (1 Mbps still fits within 1.024 MHz of bandwidth).

Chapter 11: NOAA APT Weather Satellites

NOAA's polar-orbiting weather satellites (NOAA-15, NOAA-18, NOAA-19) transmit real-time images of the Earth below them using the APT (Automatic Picture Transmission) protocol. The satellites pass overhead every 100 minutes or so, and a single pass produces a 12-minute strip image of the Earth about 2400 pixels wide and 1500 pixels tall. The console decodes these images live as the satellite passes overhead.

NOAA Satellites and Frequencies

Satellite	Frequency	Status	Pass Frequency
NOAA-15	137.620 MHz	Active	Every ~100 min
NOAA-18	137.9125 MHz	Active	Every ~100 min
NOAA-19	137.100 MHz	Active	Every ~100 min

These satellites are in sun-synchronous polar orbits at about 850 km altitude. Each pass overhead lasts 12-15 minutes and covers a swath about 2900 km wide. To find when a satellite will pass over your location, use the satellite tracker at n2yo.com or the free Android app "ISS Detector" (which also tracks NOAA satellites).

APT Signal Structure

APT transmits two image channels simultaneously as an AM-modulated 2.4 kHz subcarrier on the 137 MHz carrier:

Channel	Description	Typical Use
Channel A	Visible light (0.58-0.68 um)	Daytime cloud cover
Channel B	Thermal IR (10.5-11.5 um)	Day/night temperature

Each image line is 2080 pixels wide, transmitted at 2 lines per second (0.5 seconds per line). The line structure is:

Segment	Pixels	Content
Sync A	39	7 cycles of black-white-black at 1040 Hz
Space A	0	Unused gap
Channel A	909	Visible light image data
Sync B	39	Same as Sync A
Space B	0	Unused gap
Channel B	909	Thermal IR image data
Telemetry	127	Calibration wedges

How to Receive APT

APT reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. Tune to one of the NOAA frequencies above (e.g. 137.500 MHz for NOAA-19). The app automatically switches to WFM mode and 180 kHz bandwidth.
3. A suitable antenna (see below). The stock whip works marginally; a V-dipole or quadrifilar helix (QFH) is much better.
4. Wait for a satellite pass. The decoder searches for the sync pattern and only commits lines when it finds a valid sync - so during a real pass, the image grows line-by-line in the panel.
5. Click AUDIO ON if you want to hear the characteristic "tick-tick-tick" of the APT signal (it sounds like a slow fax machine).

Antenna Recommendations

NOAA satellites are in polar orbits, so they pass overhead from horizon to horizon. You need an antenna with broad reception pattern - a vertical whip will work but only for high-elevation passes. The best antennas for APT are:

Antenna	Difficulty	Cost	Performance
Stock whip (vertical)	Easy	Included	Poor - only high passes
V-dipole (inclined)	Easy DIY	\$5	Good for most passes
Quadrifilar helix (QFH)	Medium DIY	\$15-25	Excellent - full pass
Turnstile (crossed dipoles)	Medium DIY	\$10-20	Good - circular polarization
Discone	Buy	\$30-60	Good - broadband, also works for ADS-B

Tip: The V-dipole is the easiest improvement: build two 53 cm wires, mount them in a V shape at 120 degrees apart, with the open end pointing up at about 30 degrees from horizontal. Connect one wire to the center conductor and the other to the shield of your coax. Total cost: under \$5.

Interpreting the Image

The APT image grows line-by-line in the panel as the satellite passes. During the first few seconds, you will see the sync pattern (alternating black and white bars) as the decoder locks on. Once locked, the image data appears:

Channel A (visible light, top half)

Shows cloud cover and surface features during daytime passes. White areas are clouds (high albedo); dark areas are open water or land. Coastlines are visible when there is no cloud cover. This channel only works during daylight - at night, the channel A image is black.

Channel B (thermal IR, bottom half)

Shows temperature: bright areas are cold (high clouds, snow), dark areas are warm (ocean, low land). This channel works day and night. High-altitude clouds (cumulonimbus tops, jet streams) appear as bright white because they are very cold (around -50 to -70 degrees C at the tropopause).

What to Expect

A typical pass produces an image about 2400 pixels wide and 1500 pixels tall, covering a swath of about 2900 km. The resolution is about 4 km per pixel at the satellite's nadir (directly below). At the edges of the pass, the resolution degrades due to the slant range.

If the image appears slanted or curved, your PPM correction is off. Adjust the PPM slider in the RF & Audio panel until the image straightens out. Once calibrated, the PPM value is constant for all frequencies.

Saving the Image

The image data accumulates in the panel during the pass. To save it, right-click the image and select "Save Image As" - your browser will save the current state as a PNG file. For best results, wait until the satellite sets below the horizon (the image stops growing) before saving.

Chapter 12: Meteor M2 LRPT

Meteor M2 is a series of Russian weather satellites that transmit digital images using the LRPT (Low Rate Picture Transmission) protocol. Unlike NOAA's analog APT, LRPT transmits compressed digital images at 72 kbps, providing higher resolution (12-bit pixels vs 8-bit) and three spectral channels instead of two. The console decodes the QPSK signal and reassembles CCSDS CADU frames, which can be downloaded for offline decompression.

Meteor M2 Satellites

Satellite	Frequency	Status
Meteor M2-2	137.100 MHz	Active (launched 2019)
Meteor M2-3	137.900 MHz	Active (launched 2023)

Like NOAA, Meteor satellites are in polar orbits at about 820 km altitude, with passes lasting 12-15 minutes. The orbital inclination is slightly different from NOAA, so the two constellations often pass at different times - useful for getting more frequent imagery.

LRPT vs APT

The key differences between LRPT and NOAA APT:

Feature	NOAA APT	Meteor M2 LRPT
Modulation	AM (analog)	QPSK (digital)
Bit rate	N/A (analog)	72 kbps
Image depth	8-bit grayscale	12-bit grayscale
Channels	2 (visible + IR)	3 (visible + 1.6 um + 10.8 um)
Compression	None (line scan)	JPEG-LS compressed
Resolution	~4 km/pixel	~1 km/pixel
Frame structure	Continuous line scan	CCSDS CADU frames

Console Limitations

The console decodes the QPSK signal and reassembles CCSDS CADU frames, but does NOT decompress the image data. Decompression requires JPEG-LS, which is a specialized lossless/near-lossless compression standard that would require porting a C library to JavaScript or WebAssembly. Instead, the console accumulates raw CADU bytes and lets you download them for offline processing with tools like SatDump or the open-source meteor-decode software.

How to Receive Meteor M2

Meteor M2 reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. Tune to 137.100 MHz (Meteor M2-2) or 137.900 MHz (Meteor M2-3). The console automatically detects the band and shows the Meteor panel.
3. The same V-dipole or QFH antenna used for NOAA APT works for Meteor (both are at 137 MHz).
4. Wait for a satellite pass. The panel shows sync lock status, EVM (Error Vector Magnitude) as a signal quality metric, and accumulated frame count.
5. When frames are being received, click DOWNLOAD CADU to save the raw bytes to a .bin file for offline decompression.

Interpreting the Display

CADU Frame Count

Total CCSDS CADU frames decoded since the decoder started. Each frame is 1024 bytes. A full pass typically produces 5000-10000 frames (5-10 MB of data).

EVM (Error Vector Magnitude)

A measure of signal quality - the average distance between received QPSK symbols and their ideal positions. Lower is better: under 20% is good, under 10% is excellent. Above 40%, the decoder will struggle to maintain sync.

Bytes Received

Total raw bytes accumulated in the rolling buffer (1 MB max). When you click DOWNLOAD CADU, this buffer is saved to a file.

Post-Processing

To decompress the downloaded CADU data into viewable images, use one of these tools:

Tool	Platform	URL
SatDump	Cross-platform	github.com/SatDump/SatDump
Meteor Decoder	Linux/Mac	github.com/altusa/meteor_decoder
LRPT-Decoder	Windows	github.com/dbdexter/lrpt_decoder

SatDump is the most polished option - it accepts the raw .bin file and produces PNG images with the three spectral channels. The decompression takes a few seconds on a modern CPU.

Chapter 13: GOES HRIT

GOES (Geostationary Operational Environmental Satellite) is the US weather satellite system that provides continuous coverage of the Western Hemisphere from geostationary orbit. The GOES-R series satellites (GOES-16 East and GOES-17/18 West) transmit HRIT (High Rate Information Transmission) signals containing full-disk Earth images every 3 hours, regional images every 15 minutes, text bulletins, and EMWIN weather data.

GOES Satellites and Frequencies

Satellite	Frequency	Location	Coverage
GOES-16 (East)	1685.7 MHz	75.2 W	Atlantic, Americas
GOES-17/18 (West)	1694.1 MHz	137.2 W	Pacific, Western US

GOES satellites are geostationary, meaning they appear fixed in the sky at about 35,786 km altitude. Once you point your antenna at them, you can receive 24/7 - no waiting for satellite passes.

HRIT Signal Structure

HRIT transmits at 927 kbps using BPSK modulation with the following frame structure:

Component	Size	Description
Sync word	32 bits	0x1ACFFC1D (CCSDS standard)
CADU frame	1024 bytes	Channel Access Data Unit
Viterbi coding	Rate 1/2, K=7	Forward error correction
Reed-Solomon	RS(255, 223)	Additional error correction

Files are assembled from multiple CADU frames. The HRIT primary header (first 10 bytes after the sync) indicates the file type, total file size, and a unique file name. The console parses these headers to track ongoing file reception.

File Types

GOES HRIT transmits several file types:

Type	Description	Frequency
Image	Full-disk or regional Earth image	Every 3 hours (full disk)
Text	Weather bulletins and forecasts	Variable
JPEG-LS	Compressed image data	Variable
EMWIN	Emergency Managers Weather Info	Continuous
DCS	Data Collection System (sensor data)	Variable

Antenna Requirements

GOES HRIT is a weak signal that requires a directional antenna and a low-noise amplifier (LNA). The stock RTL-SDR whip will NOT work. Recommended antennas:

Antenna	Gain	Difficulty	Cost
10-turn helical	13-15 dBi	Medium DIY	\$30-50
60cm dish + patch feed	20+ dBi	Hard	\$80-150
Commercial GOES dish	24 dBi	Buy	\$200-400

The antenna must be pointed at the satellite with an accuracy of about 5 degrees. For a helical, this means manual aiming based on the satellite's azimuth and elevation from your location (use an online satellite look-angle calculator). For a dish, a small motorized positioner is helpful.

Warning: An LNA (low-noise amplifier, ~\$20-30) is essential. Place it directly at the antenna feed - before any coax - to overcome coax losses. Without an LNA, you will not receive anything. The RTL-SDR's internal noise figure is too high for the weak GOES signal.

How to Receive GOES HRIT

GOES HRIT reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. A directional antenna (helical or dish) with LNA, pointed at the satellite.
3. Tune to 1685.7 MHz (GOES-East) or 1694.1 MHz (GOES-West).
4. Set sample rate to 2.4 Msps, gain to maximum, AGC off.
5. The GOES panel appears automatically when tuned to 1680-1700 MHz.

Interpreting the Display

Frame Count and Bytes

Shows the total CADU frames decoded and the total bytes accumulated. A healthy GOES signal produces 800-1000 frames per second (927 kbps / 1024 bytes per frame). If you are not getting frames, your antenna is not pointed correctly or the LNA is not powered.

Bit Error Rate (BER)

After Viterbi decoding, the bit error rate should be under 1% for reliable file reception. Higher BER indicates a weak signal or mis-pointed antenna. The Reed-Solomon FEC can correct up to 16 byte errors per 255-byte codeword, so even with a moderate BER you can still receive files correctly.

Current File

When a file is being received, the panel shows the file type, a progress bar indicating how many bytes have been received out of the total, and the expected total size. Full-disk Earth images are typically 8-12 MB and take 15-20 minutes to receive.

Completed Files

Lists up to 20 most recently completed files with type, size, and timestamp. These files are stored in the bridge's recordings/ directory and can be downloaded via the HTTP server (see Chapter 6).

Post-Processing

GOES HRIT image files use a custom image format that needs to be decoded into viewable PNGs. The goestools package (Linux/macOS) or SatDump can do this conversion. The decompressed images are typically 1080p grayscale (single channel) or false-color composites of multiple spectral bands.

Chapter 14: POCSAG Pagers

POCSAG (Post Office Code Standardization Advisory Group) is the standard protocol for one-way pagers, still widely used by hospitals, fire departments, emergency services, and IT alerting systems. Despite the rise of smartphones, pagers remain popular in environments where cellular service is unreliable or where messages must reach recipients instantly without acknowledgement delays. The console decodes POCSAG messages in real time.

POCSAG Signal Structure

POCSAG transmits at three standard baud rates (512, 1200, and 2400 bps) using 2-FSK (Frequency Shift Keying) modulation with ± 4.5 kHz deviation. The frame structure is:

Component	Size	Description
Preamble	576 bits	Alternating 1-0 pattern for synchronization
Sync word	32 bits	0x7CD215D8 (frame sync)
Batches	16 codewords	1 address + 15 data, or all data
Codeword	32 bits	1 bit type + 18 bits address/data + 10 BCH + 1 parity

Frequencies

POCSAG pagers operate on several bands:

Band	Frequency Range	Common Users
US Pager Band	929-932 MHz	Commercial, hospital, fire/EMS
VHF	138-174 MHz	Some fire/EMS, business
UHF	440-470 MHz	IT alerting, hospitals

Message Types

POCSAG messages come in three types, indicated by the function code bits:

Type	Code	Description	Encoding
Numeric	0	Numeric-only message (phone numbers, codes)	BCD, 5 digits per codeword
Tone	1	Beep only (no message content)	Empty
Alphanumeric	2 or 3	Text message (ASCII)	7-bit ASCII, 3 chars per codeword

How to Receive POCSAG

POCSAG reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. Tune to a pager frequency. The US pager band (929-932 MHz) is most active. The console auto-detects the band and shows the Messages panel with the Pagers tab selected.
3. The stock RTL-SDR whip works for nearby pager towers. For better range, a 900 MHz Yagi is recommended.
4. Set demod mode to NFM with 25 kHz bandwidth.
5. Watch the Messages panel - decoded messages appear in real time with the pager address (the destination pager ID), function code, and message text.

Interpreting Messages

Pager Address

Each pager has a unique 7-digit address (sometimes called a CAP code or RIC). Messages are addressed to specific pagers, and only the addressed pager will display the message. The console shows the address of each received message.

Numeric Messages

Numeric messages are typically phone numbers (the recipient is expected to call back) or short numeric codes (e.g. 911 for emergency, 4 for "call the office"). The BCD encoding uses a special character set that includes U (empty), -,), and (in addition to digits 0-9.

Alphanumeric Messages

Alphanumeric messages are 7-bit ASCII text. They can be up to several hundred characters long, requiring multiple codewords. Common examples include dispatch messages ("Engine 5 respond to 123 Main St for structure fire"), IT alerts ("Server SRV-01 CPU at 95%"), and hospital pages ("Dr. Smith call extension 4521").

Warning: Pager messages may contain sensitive personal or medical information. Be aware of privacy considerations when sharing or storing decoded messages. In some jurisdictions, intercepting pager traffic may be subject to legal restrictions. Check your local laws.

Performance Notes

The POCSAG decoder supports all three baud rates (512, 1200, 2400 bps) and automatically detects the rate by trying all three. The decoder uses a 1-bit error tolerance on the sync word to handle noisy signals. BCH parity checking is not strictly enforced (we tolerate some false positives in exchange for catching marginal messages).

In a typical urban area, you can expect 5-50 messages per minute during peak hours. Rural areas may produce only a few messages per hour. The decoder keeps the last 100 messages in the panel - older messages are dropped.

Chapter 15: ACARS Aircraft Messaging

ACARS (Aircraft Communications Addressing and Reporting System) is a digital short-message system used by airlines to exchange text messages between aircraft and dispatchers. It operates on VHF (131.55 MHz primary, plus 131.725 and 131.825 MHz) and via satellite (Inmarsat). The console decodes the VHF ACARS signals, providing real-time visibility into aircraft-to-ground communications.

ACARS Signal Structure

ACARS transmits at 2400 bps using MSK (Minimum Shift Keying) modulation, which is a special case of FSK with deviation exactly half the bit rate. The bit stream is NRZI encoded (Non-Return-to-Zero Inverted: a 0 bit is a transition, a 1 bit is no transition). The frame structure is:

Field	Size	Description
Bit sync	16 bits	Alternating 1-0 pattern
Preamble	Variable	Frame start marker
Mode	1 byte	1=downlink, 2=uplink
Flight ID	6 bytes	Flight identifier (e.g. UAL123)
Aircraft reg	7 bytes	Tail number (e.g. N123AB)
Message label	2 bytes	Message type code
Message number	4 bytes	Sequence number (e.g. M01A)
Message text	Variable	Free-form ASCII text
ETX	1 byte	End-of-text marker

Message Labels

ACARS messages are categorized by 2-character labels:

Label	Description
H1	Position report (routine)
H2	Position report (request)
Q0	ACARS system message
Q1	Cancel downlink
Q3	Oceanic request
10	Weather request
11	Weather data
15	Free text message

Label	Description
13	Engine data
5U	Out/Off/On/In (OOOI) report

How to Receive ACARS

ACARS reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. Tune to 131.55 MHz (primary ACARS channel). The console auto-detects and shows the Messages panel with the ACARS tab selected.
3. Set demod mode to AM with 12 kHz bandwidth.
4. The stock RTL-SDR whip works for ACARS if you are within 50 miles of an airport or major flight path. For better range, a VHF airband antenna is recommended.
5. Watch the Messages panel - ACARS messages appear in real time with the flight ID, aircraft registration, message label, and text body.

Interpreting Messages

OOOI Reports

OOOI stands for Out, Off, On, In - the four phases of flight. ACARS sends an OOOI message at each phase transition: Out (pushback from gate), Off (wheels up), On (wheels down at destination), In (arrival at gate). These messages let airlines track flight progress automatically.

Position Reports

Aircraft send periodic position reports (label H1) containing latitude, longitude, altitude, and time. These are typically sent every 5-10 minutes during cruise. Some airlines send them more frequently over oceanic routes.

Free Text Messages

Free text messages (label 15) are used for ad-hoc communication between pilots and dispatchers. Common examples include weather diversions, gate assignments, mechanical issues, and passenger requests.

Realistic Expectations

ACARS activity is highest near major airports and on busy air routes. In a typical urban area within 50 miles of a major airport, you can expect 10-50 messages per hour. Rural areas away from flight paths may produce only a few messages per day. Messages are typically short (50-200 characters) and arrive in bursts when multiple aircraft are in range.

Tip: Pair ACARS decoding with ADS-B (Chapter 10) for a complete aircraft monitoring setup. ADS-B shows aircraft positions on the radar plot, while ACARS shows the text messages each aircraft is exchanging with dispatchers.

Chapter 16: Inmarsat STD-C

Inmarsat-C is a store-and-forward messaging service used by maritime vessels, remote assets, and aviation for low-bandwidth text communication. The satellites are geostationary at L-band (1537-1545 MHz), and the system supports marine distress alerts (GMDSS), ship-to-shore email, and asset tracking. The console decodes the STD-C (Standard-C) TDM channel, which is the always-on broadcast from the Network Coordination Station.

Inmarsat Satellites

Satellite	Position	Frequency	Coverage
I-3 AOR-E	15.5 W	1543.5 MHz	Atlantic East, Europe, Africa
I-3 AOR-W	54 W	1543.5 MHz	Atlantic West, Americas
I-3 IOR	64 E	1537.5 MHz	Indian Ocean, Asia, Australia
I-3 POR	178 E	1537.5 MHz	Pacific Ocean

Each satellite is in geostationary orbit at about 35,786 km altitude. The satellite you can receive depends on your location - find the nearest one and point your antenna at it.

STD-C Signal Structure

STD-C transmits at 1200 bps using BPSK modulation with the following frame structure:

Component	Size	Description
Frame duration	8.64 seconds	One TDM frame
Total bits	10,368	Per frame
Channels	12	Per frame
Channel size	864 bits	108 bytes per channel
Bulletin board	Channel 1	NCS ID + LES assignments
Convolutional code	Rate 1/2, K=7	Forward error correction

Antenna Requirements

Inmarsat STD-C is a weak L-band signal requiring a directional antenna with circular polarization and an LNA. The stock RTL-SDR whip will NOT work. Recommended antennas:

Antenna	Gain	Difficulty	Cost
Patch antenna (RHCP)	6-9 dBi	Easy DIY	\$15-25
Helical (4-turn)	10-12 dBi	Medium DIY	\$25-40
Helical (8-turn)	13-15 dBi	Medium DIY	\$35-50

Antenna	Gain	Difficulty	Cost
Commercial Inmarsat antenna	10-15 dBi	Buy	\$80-200

The antenna must be pointed at the satellite with an accuracy of about 10 degrees. Use an online satellite look-angle calculator to determine azimuth and elevation from your location. An LNA (~\$20-30) is essential - place it directly at the antenna feed, before any coax.

How to Receive Inmarsat STD-C

STD-C reception requires:

1. A real RTL-SDR V3 connected (see Chapter 5).
2. A directional antenna (patch or helical) with LNA, pointed at the satellite.
3. Tune to 1537.5 MHz (IOR/POR) or 1543.5 MHz (AOR-E/AOR-W). The Inmarsat panel appears automatically.
4. Set sample rate to 2.4 Msps, gain to maximum, AGC off.
5. The panel shows NCS ID, active LES list, and any decoded messages.

Interpreting the Display

NCS ID

The Network Coordination Station ID. There is one NCS per satellite region (typically 4 NCS stations globally). The NCS broadcasts the bulletin board that announces which Land Earth Stations (LES) are active.

LES List

Land Earth Stations are the ground stations that route messages to and from the satellites. Each LES has a unique 4-digit hex ID. Active LES IDs are announced in the bulletin board. When a message is sent from a vessel, the LES that handles it is logged in the message header.

Messages

Decoded messages are listed with the sender (LES ID), timestamp, and text content. Common message types include:

- GMDSS distress alerts (maritime emergencies)
- Maritime safety information (MSI)
- Fleet management messages
- Asset tracking pings
- Email gateway messages

Warning: Inmarsat STD-C messages may contain sensitive maritime safety or commercial information. Be aware of privacy considerations when sharing decoded messages. GMDSS distress alerts should be reported to authorities if received.

Realistic Expectations

STD-C activity depends on the satellite you can receive and the time of day. AOR-E and IOR are typically busiest (covering Europe and Asia respectively). Expect 1-10 messages per hour in a 24-hour period - much lower than ACARS or POCSAG. The TDM channel itself is always broadcasting (you will see the NCS ID quickly), but actual messages are sparse.

Chapter 17: GPS L1 C/A Decoder

The Global Positioning System broadcasts civil signals on the L1 frequency (1575.42 MHz) using CDMA (Code Division Multiple Access) spread-spectrum modulation. Each of the 32 GPS satellites broadcasts a unique 1023-chip Gold code at 1.023 Mchips/second, with a 50 bps navigation message on top. The console generates the C/A codes for PRN 1-12 and attempts to acquire and track each satellite, showing signal strength (C/N0) and pseudoranges.

GPS Signal Structure

The L1 C/A (Coarse/Acquisition) signal has the following characteristics:

Parameter	Value
Carrier frequency	1575.42 MHz (L1)
Code rate	1.023 Mchips/s
Code length	1023 chips (1 ms repeat)
Code type	Gold code (PRN 1-32)
Modulation	BPSK on spread code
Nav message rate	50 bps
Nav message length	25 frames x 5 subframes x 10 bits = 12.5 min
Satellites in orbit	31 active (PRN 1-32, with gaps)

C/A Code Generation

Each GPS satellite's C/A code is generated by XORing two LFSRs (Linear Feedback Shift Registers) called G1 and G2:

G1 polynomial: $x^{10} + x^3 + 1$ (taps at positions 3 and 10)

G2 polynomial: $x^{10} + x^9 + x^8 + x^6 + x^3 + x^2 + 1$ (taps at 2, 3, 6, 8, 9, 10)

Each satellite uses a different pair of G2 taps (the "phase selection") to produce its unique Gold code. For example, PRN-1 uses taps 2 and 6, PRN-2 uses taps 3 and 7, and so on. The full table of 32 tap pairs is defined in ICD-GPS-200C. The console implements all 12 most active PRNs.

Acquisition Process

To acquire a GPS satellite, the decoder must find two unknowns simultaneously:

Doppler Shift

GPS satellites move at about 3.9 km/s, causing Doppler shifts of up to ± 5 kHz on the L1 carrier. The decoder searches Doppler in 500 Hz steps, mixing the received signal down by each candidate Doppler frequency.

Code Phase

The 1023-chip C/A code repeats every 1 ms. The decoder must find the exact code phase (which chip is being received at which moment). This is done by correlating the received signal against the local code at each of 1023 possible code phases.

When both Doppler and code phase match, the correlation produces a sharp peak. If the peak exceeds a threshold (corresponding to about 30 dB-Hz C/N0), the satellite is acquired and tracking begins.

Tracking

Once acquired, the decoder tracks the satellite by continuously correlating the received signal with the local code at the tracked Doppler and code phase. Small adjustments keep the correlation peak centered as the satellite moves. The integrated correlation over 20 ms produces one nav message bit (50 bps).

Antenna Requirements

GPS signals are extremely weak (-130 dBm at the Earth's surface) and require an active antenna with built-in LNA. The stock RTL-SDR whip will NOT work. Recommended antennas:

Antenna	Type	Cost	Notes
Active GPS patch	MCX connector	\$5-15	Standard, works well
Active GPS ceramic	SMA connector	\$8-20	Smaller, similar performance
External GPS antenna	Roof-mounted	\$20-50	Best for indoor receivers

Most active GPS antennas have an MCX or SMA connector that plugs directly into the RTL-SDR V3's antenna jack (with an adapter if needed). The antenna is powered by DC injected onto the coax (typically 3-5V). The RTL-SDR V3 has a software-controlled bias tee that can provide this - check the Hardware Source panel for a "Bias Tee" option (if not available, you need an external bias tee).

Warning: Without a clear view of the sky, GPS reception is very poor. Indoors you may get 0-2 satellites with weak signal. Outdoors with a clear sky view, you can expect 6-12 satellites at good strength.

How to Receive GPS

GPS reception requires:

1. A real RTL-SDR V3 connected with an active GPS antenna (see above).
2. Tune to 1575.42 MHz. The GPS panel appears automatically when tuned to 1570-1580 MHz.
3. Set sample rate to 2.4 Msps (default).
4. Set gain to maximum, AGC off.
5. The panel shows up to 12 satellites (PRN 1-12) with their tracking status, C/N0, Doppler shift, and pseudorange.

Interpreting the Display

Per-PRN Status

Each satellite shows one of three states:

State	Description
SRCH	Searching - correlating against the satellite's C/A code
TRK	Tracking - acquired and tracking, decoding nav bits

C/N0

Carrier-to-noise density ratio in dB-Hz. This is the standard GPS signal strength metric. Values: 30+ dB-Hz = usable, 35+ dB-Hz = good, 40+ dB-Hz = excellent. Below 28 dB-Hz, the signal is too weak to track.

Doppler Shift

The Doppler shift in Hz, positive or negative. This changes as the satellite moves across the sky. Range is ± 5000 Hz for typical GPS satellites.

Pseudorange

An estimate of the distance to the satellite in megameters (millions of meters). GPS satellites orbit at about 20,200 km, so pseudoranges should be around 20,000-25,000 Mm. The pseudorange is computed from the code phase: $(\text{code phase} / 1.023 \text{ Mchips/s}) * \text{speed of light}$.

Position Fixing

A full GPS position fix requires 4+ satellites with valid ephemeris data (satellite positions decoded from the nav message) and trilateration. The console does NOT compute a position fix - it tracks satellites and shows signal strength + pseudoranges, but does not parse the ephemeris or perform trilateration. This is left as a future enhancement. To get an actual position fix, use a dedicated GPS app or library.

Realistic Expectations

With a proper active GPS antenna and clear sky view, you can expect to acquire 6-12 satellites within 30-60 seconds. Without a clear sky view (indoors, near buildings), you may only acquire 0-3 satellites. The decoder searches for satellites continuously, so newly risen satellites are acquired automatically.

Chapter 18: Scanner Mode

The scanner automates signal discovery. Instead of manually clicking through frequencies looking for activity, the scanner steps through a band (or watches the current spectrum) and either tunes to the strongest signal or collects all signals into a list. Three scan modes cover different use cases.

Scan Modes

Peak Scan

Continuously finds the strongest signal in the current spectrum view and tunes to it every 800 ms. This is the simplest mode - useful for finding whatever is loudest right now. It is great for FM broadcast (tune to the strongest local station) or for monitoring a band where you expect one strong signal at a time.

Click the "FIND STRONGEST SIGNAL" button to start peak scanning. The scanner retunes the SDR to the peak every 800 ms and updates the spectrum display to show the new center frequency. Click the button again (now labeled "STOP PEAK SCAN") to stop.

Sweep Scan

Sweeps through a band in 70 percent sample-rate steps, collecting every signal above -60 dBFS into a sorted list. After the sweep completes, the list shows each found signal with frequency and strength, sorted strongest first. Click any entry to tune back to it.

To sweep a band, pick a band preset (FM Broadcast, Airband, 2m Ham, NOAA, Marine, 70cm Ham, GMRS/FRS, Shortwave, or CB Radio) and click SWEEP. The scanner tunes across the band, pausing 120 ms at each step to capture the spectrum. A full FM broadcast sweep (87.5-108 MHz) takes about 30 seconds at 2.4 Msps.

Squelch Scan

Steps through a band in channel-sized increments, stopping for 3 seconds on each frequency where the signal exceeds the squelch threshold. This is the traditional "scanner radio" behavior - it pauses on active channels so you can hear what is being said, then continues.

Set the squelch threshold in the RF & Audio panel (typically 15-25 percent). Pick a band preset and click SQL. The scanner tunes to each channel, waits 150 ms for the spectrum to settle, checks if the center bin exceeds the squelch threshold, and either pauses for 3 seconds (signal found) or moves to the next channel.

Band Presets

The scanner has 9 pre-configured band presets:

Band	Range	Step	Demod	Use
FM Broadcast	87.5-108 MHz	200 kHz	WFM	Local FM stations
Airband (AM)	118-137 MHz	25 kHz	AM	ATC communications
2m Ham Band	144-148 MHz	12.5 kHz	NFM	Ham radio voice

Band	Range	Step	Demod	Use
NOAA Weather	162.4-162.55 MHz	25 kHz	NFM	Weather radio
Marine VHF	156-162 MHz	25 kHz	NFM	Boat communications
70cm Ham	420-450 MHz	25 kHz	NFM	Ham radio voice
GMRS / FRS	462-467 MHz	12.5 kHz	NFM	Family radio
Shortwave (HF)	3-30 MHz	5 kHz	AM	International broadcast
CB Radio	26.965-27.405 MHz	10 kHz	AM	Citizens Band

Each preset configures the appropriate demodulator mode, bandwidth, and step size automatically. The step size matches the channel spacing used in that band - for example, marine VHF uses 25 kHz channels, so the scanner steps by 25 kHz to land on each valid channel frequency.

Interpreting Results

Found Signals List

After a sweep or squelch scan, the found signals list shows each detected frequency with its signal strength in dBFS. The list is sorted strongest first. Click any entry to tune back to it. The list persists until you click Clear or start a new scan.

Live Position Indicator

During a scan, the panel shows the current scanning frequency in monospace text at the bottom. This updates in real time so you can watch the scanner's progress through the band.

Performance Notes

Sweep scan moves in 70 percent sample-rate steps to overlap coverage and avoid missing signals at the edges of each view. At 2.4 Msps, each step covers 1.68 MHz, and a full FM broadcast sweep (20.5 MHz wide) takes about 12 steps - roughly 30 seconds including the 120 ms pause at each.

Squelch scan is slower because it pauses on active channels. In a busy band (like airband near an airport), a full 19 MHz scan might take 5-10 minutes because the scanner stops on every active frequency. Use peak scan for quick discovery, sweep scan for band mapping, and squelch scan for listening to conversations.

Chapter 19: Notch Filter & Interference

Strong local signals can swamp your RTL-SDR's front end, making it impossible to hear weaker signals on nearby frequencies. The most common example is a strong local FM broadcast station that desensitizes the receiver, preventing reception of weaker signals across the entire FM band. The notch filter lets you suppress these interferers by removing their IQ content before demodulation.

How Notch Filtering Works

A notch filter is a biquad IIR (infinite impulse response) filter that places zeros on the unit circle at the notch frequency (creating a deep null) and poles just inside the unit circle (to keep the notch narrow). The result is a narrow, deep suppression at a specific frequency, without affecting the rest of the spectrum.

The console applies notch filters to the complex IQ stream before demodulation. Each notch is applied independently to the I and Q channels. Up to 16 notches can be active simultaneously. The filter is computed in real time, adding about 5 percent CPU load per notch on a modern machine.

Manual Notches

To add a manual notch, enter the frequency offset in Hz (relative to the tuned center frequency) in the input field and click the + button. For example, if you are tuned to 95.5 MHz and a strong local station is at 96.5 MHz, the offset is +1,000,000 Hz (1 MHz). The notch appears in the active list with an amber badge indicating it is manual.

Manual notches persist until you remove them (click the X next to the notch) or change the tuned frequency significantly. They are useful for persistent interferers like local FM stations, paging towers, or pager transmitters.

Auto-Detect Mode

Toggle the "Auto-detect interferers" switch to enable automatic notch detection. In auto-detect mode, the console scans the spectrum every 100 ms and looks for bins whose power is more than 25 dB above the local noise floor. Any such bin is automatically added as a notch, with a minimum spacing of 50 kHz to avoid duplicate notches on the same signal.

Auto-detected notches appear with a green AUTO badge. They are recomputed every 100 ms, so they adapt to changing interference. Click "CLEAR AUTO" to remove all auto-detected notches at once (manual notches are preserved).

Tip: Auto-detect is best for quickly identifying interferers you did not know about. Once you have identified a persistent interferer (like a local FM station), convert it to a manual notch by removing the auto notch and adding it manually with the + button. Manual notches are stable across frequency changes.

When to Use Notches

Strong Local FM Station

If you live near an FM broadcast tower, the strong signal (often 0 dBFS or higher on the spectrum) can desensitize the RTL-SDR across the entire FM band. Add a manual notch at the offset of the offending station (e.g. if you are at 95.5 MHz and the station is at 96.5 MHz, add a notch at +1,000,000 Hz). The notch suppresses the strong signal, allowing weaker stations to be heard.

Pager Tower Nearby

If you live near a pager tower (929-932 MHz), the constantly-transmitting pager signal can interfere with reception of weaker signals in the 900 MHz band. Add a manual notch at the pager tower's frequency offset.

ADS-B Reception

ADS-B at 1090 MHz is sometimes interfered with by nearby 1090 MHz sources (rare, but possible). If your ADS-B decode rate is unexpectedly low, check the spectrum for any strong signals within ± 10 MHz of 1090 MHz and notch them.

What Notches Cannot Do

Notch filters suppress specific frequencies but cannot fix fundamentally weak signal reception. If your antenna is poorly positioned or the signal you want is below the noise floor, notches will not help - you need a better antenna. Also, notches cannot fix interference that is broadband (covering many frequencies) or that comes from inside your own equipment (e.g. USB 3.0 interference).

USB 3.0 ports generate broadband interference around 2.4 GHz that can interfere with SDR reception. If you see a high noise floor across the entire spectrum, try plugging the RTL-SDR into a USB 2.0 port instead, or use a USB extension cable to move the dongle away from the computer.

Q Factor

The Q factor controls how narrow the notch is. Higher Q = narrower notch (deeper suppression of a single frequency, less affect on adjacent frequencies). Lower Q = wider notch (suppresses a range of frequencies). The console uses a default Q of 30, which gives a notch about 1 kHz wide at 100 kHz offset. For most applications, the default Q is fine. Manual notches let you adjust Q if needed.

Chapter 20: Recording & Playback

The console supports two types of recording: IQ recording (raw RF data, written to disk on the bridge) and audio WAV recording (demodulated audio, captured in the browser). Both produce downloadable files in standard formats compatible with other SDR software.

IQ Recording

What it Records

IQ recording captures the raw IQ stream from the SDR, exactly as received from rtl_tcp. Each sample is two bytes (unsigned 8-bit I + unsigned 8-bit Q), giving a data rate of $\text{sample_rate} * 2$ bytes per second. At 2.4 Msps, that is 4.8 MB/s - so a 5-minute recording is about 1.4 GB.

How to Record

1. Connect to real hardware (see Chapter 5).
2. Tune to the frequency you want to record.
3. Click the RECORD button in the IQ Recording panel (right column). The button turns red and shows a pulsing indicator.
4. Live stats appear: duration (seconds), file size (MB), and write rate (MB/s).
5. Click STOP when done. The recording is saved on the bridge in the `recordings/` directory.

File Format

Files are saved as raw unsigned 8-bit IQ (.raw), with no header. The filename includes the timestamp, frequency, and sample rate: e.g. `iq-2026-01-15T12-34-56-91500000MHz-2400ksps.raw`. The format is compatible with SDR#, GQRX, GNU Radio, and most other SDR software.

Downloading

After stopping a recording, it appears in the saved recordings list in the panel. Click the download icon next to any recording to download it via HTTP (the bridge serves files on port 8081). Files are listed with size, timestamp, and download link.

Playback in Other Software

To replay a recording in SDR#: select File -> Play, choose "Raw IQ File", set the sample format to "8-bit unsigned", and enter the original sample rate (from the filename). In GNU Radio, use a File Source block with item type "complex", then multiply by 1/128 and subtract 1 to convert to float.

Audio WAV Recording

What it Records

Audio WAV recording captures the demodulated audio - exactly what you hear through your speakers. The audio is encoded as 16-bit PCM mono WAV at the demodulator's output rate (48 kHz for WFM/NFM/AM, 24 kHz for SSB/CW).

How to Record

1. Connect to real hardware and enable audio (AUDIO ON).
2. Click the RECORD AUDIO button in the Audio Recorder panel (right column, below the IQ Recording panel).
3. Live duration counter shows elapsed recording time.
4. Click STOP when done. The clip is encoded as a WAV file and added to the saved clips list.

Downloading

Saved clips appear in the clips list with duration, file size, download link, and delete button. Click the download icon to save the WAV file to your Downloads folder. Filenames include timestamp, frequency, and demod mode: e.g. audio-2026-01-15T12-34-56-91500000MHz-WFM.wav.

Preset Sync

When connected to real hardware, the Memory Bank panel (right column) shows two cloud icons in the header: download (pull presets from bridge) and upload (push presets to bridge). This lets you sync your bookmarks across devices - save bookmarks on your desktop, push them to the bridge, then pull them on your laptop or phone.

Push to Bridge

Click the cloud-up icon to save your current bookmarks to the bridge. The bridge writes them to presets.json in its working directory. The status message shows how many bookmarks were saved.

Pull from Bridge

Click the cloud-down icon to load bookmarks from the bridge. Your local bookmarks are replaced with the bridge's set. The status message shows how many bookmarks were loaded.

Tip: Use preset sync to maintain a shared bookmark list across multiple devices. For example, set up bookmarks on your desktop at home, push them to the bridge, then pull them on your laptop when you take it to a different location.

Chapter 21: Antenna Guide

The antenna is the most important component of any SDR setup - more important than the dongle itself. A \$30 dongle with a \$50 antenna will outperform a \$200 dongle with the stock whip. This chapter provides antenna recommendations for each band the console supports, plus connector and LNA guidance.

Antenna Basics

An antenna is a transducer that converts radio frequency electrical signals into electromagnetic waves (and vice versa). The key parameters are:

Parameter	Description	Why it Matters
Gain	How much the antenna focuses energy in one direction	Higher gain = more range but narrower pattern
Polarization	Orientation of the electromagnetic wave (vertical, horizontal, circular)	Must match the transmitter for best reception
Bandwidth	Range of frequencies the antenna works over	Wide bandwidth = one antenna for many bands
Impedance	AC resistance at the operating frequency (typically 50 ohms)	Must match the coax and dongle
Pattern	Direction(s) of maximum sensitivity	Omnidirectional vs directional

Antenna Recommendations by Band

FM Broadcast (87.5-108 MHz)

The stock RTL-SDR whip works adequately for strong local FM stations. For better reception of weaker or more distant stations, a half-wave dipole (about 1.5 meters long) mounted vertically or horizontally will improve signal strength by 3-6 dB. A discone antenna provides broadband coverage but less gain than a dedicated dipole.

Airband (118-137 MHz)

Airband signals are vertically polarized. The stock whip works for nearby aircraft (within 20-30 miles). For better range, a vertical dipole or ground plane antenna cut for 130 MHz is recommended. A discone antenna provides good airband reception plus broadband coverage for other bands.

NOAA APT (137 MHz)

NOAA satellites transmit right-hand circular polarization (RHCP). A V-dipole is the simplest effective antenna - build two 53 cm wires in a V shape at 120 degrees apart, inclined about 30 degrees from horizontal. For better performance, a quadrifilar helix (QFH) antenna provides full circular polarization and works for entire satellite passes (including low-elevation passes where the satellite is near the horizon).

ADS-B (1090 MHz)

ADS-B signals are vertically polarized. The stock whip works marginally. For good range (100-200 nm), a 1090 MHz antenna is recommended - either a 1/4 wave ground plane, an 8-element collinear, or a commercial 1090 MHz antenna. A low-noise amplifier (LNA) at the antenna feed can extend range to 250+ nm.

POCSAG Pagers (929-932 MHz)

Pager signals are vertically polarized. The stock whip works for nearby pager towers. For better range, a 900 MHz Yagi pointed at the nearest pager tower is recommended. A 4-element Yagi provides 8-10 dB gain over the stock whip.

GPS L1 (1575.42 MHz)

GPS signals are right-hand circular polarized (RHCP) and extremely weak. An active GPS antenna (with built-in LNA) is essential - the stock whip will NOT work. Active GPS antennas cost \$5-15 and have MCX or SMA connectors. They require DC power (3-5V) injected onto the coax - the RTL-SDR V3 can provide this via its software-controlled bias tee.

GOES HRIT (1685 MHz) / Inmarsat STD-C (1537-1545 MHz)

These L-band satellite signals are RHCP and very weak. A directional antenna (10-turn helical or 60cm dish with patch feed) plus an LNA is required. The antenna must be pointed at the satellite with about 5-degree accuracy. Use an online satellite look-angle calculator to find azimuth and elevation from your location.

Connector Types

The RTL-SDR V3 has an MCX (Micro-Coax) antenna connector. Most aftermarket antennas use SMA or BNC connectors. You will need an adapter:

From	To	Adapter Type	Cost
MCX (RTL-SDR)	SMA (most antennas)	MCX male to SMA female	\$3-5
MCX (RTL-SDR)	BNC (older antennas)	MCX male to BNC female	\$4-6
MCX (RTL-SDR)	F (TV coax)	MCX male to F female	\$3-5
SMA (active GPS)	MCX (RTL-SDR)	SMA male to MCX female	\$3-5

Use high-quality adapters (not the cheapest) - cheap adapters can introduce intermodulation distortion and signal loss. Aim for brass or nickel-plated adapters from reputable brands.

LNA (Low-Noise Amplifier)

An LNA amplifies weak signals at the antenna feed, before they travel down the coax (where they would be attenuated further). This is critical for weak signals like GPS, GOES, and Inmarsat. LNAs are rated by noise figure (NF, lower is better) and gain (typically 20-30 dB).

Place the LNA directly at the antenna feed - before any coax. A 5-meter run of cheap RG-58 coax at 1.5 GHz loses about 3 dB of signal - placing the LNA before the coax overcomes this loss. For GPS and satellite reception, an LNA with NF under 1 dB is recommended (the RTL-SDR's internal noise

figure is about 3.5 dB, so the LNA makes a real difference).

Coax Cable

Coax quality matters more at higher frequencies. For runs over 3 meters at frequencies above 1 GHz, use low-loss coax like LMR-240 or LMR-400 instead of the cheap RG-58 that comes with most antennas. The difference can be 3-6 dB of signal loss, which is the difference between a usable signal and static.

Tip: If you cannot upgrade the coax, mount the RTL-SDR dongle at the antenna and use a long active USB extension cable instead. This eliminates coax loss entirely - the IQ data travels over USB (digital, lossless) instead of RF coax (analog, lossy).

Chapter 22: Keyboard Shortcuts

The console supports keyboard shortcuts for common operations. Press the ? key (or click the keyboard icon in the bottom-right corner) to open the shortcuts help overlay at any time. Shortcuts are disabled while typing in input fields.

Complete Shortcut Reference

Key	Action	Notes
Space / M	Toggle audio mute	Same as clicking AUDIO ON/OFF
Up arrow	Tune up 25 kHz	Step size for quick scanning
Down arrow	Tune down 25 kHz	Step size for quick scanning
Left arrow	Previous demod mode	Cycles WFM - NFM - AM - USB - LSB - CW - RAW
Right arrow	Next demod mode	Cycles the other direction
[or {	Lower sample rate	Steps through 8 sample rate presets
] or }	Higher sample rate	Steps through 8 sample rate presets
- or _	Lower gain by 1 dB	Only when AGC is off
+ or =	Higher gain by 1 dB	Only when AGC is off
A	Toggle AGC	Switches between manual and auto gain
R	Toggle IQ recording	Same as clicking REC button
S	Toggle scan mode	Starts/stops peak scan
F	Toggle fullscreen spectrum	ESC to exit
?	Show keyboard shortcuts overlay	Click or press ? key
Escape	Close dialog / exit fullscreen	Closes any modal or fullscreen mode

Workflow Tips

Quick Scanning

To quickly scan a band: press S to start peak scanning (the scanner auto-tunes to the strongest signal every 800 ms), then press Space to listen. If you hear something interesting, press S again to stop scanning, then press R to record.

Demodulator Switching

When exploring an unknown band, use the left and right arrow keys to cycle through demodulator modes. Listen to each one - if the audio sounds right (voice, music, morse, etc.), you have found the correct demod for that signal.

Sample Rate Adjustment

Use the [and] keys to adjust sample rate. Lower sample rates (240 ksps, 1.024 Msps) are useful for narrow signals and reduce CPU load. Higher sample rates (2.4 Msps, 3.2 Msps) are needed for wide signals (FM broadcast, ADS-B) and provide more spectrum visibility.

Gain Tuning

With AGC off, use - and + to fine-tune gain in 1 dB steps. Watch the spectrum while adjusting - you want the noise floor around -80 to -70 dBFS and signals peaking around -30 to -10 dBFS. If you see clipping (signals hitting 0 dBFS), reduce gain.

Accessibility

All shortcuts work with both lowercase and uppercase letters (no Shift required). The keyboard help overlay is keyboard-navigable: Tab to move between sections, Enter to close. Screen readers will announce the shortcut names and descriptions.

Chapter 23: Troubleshooting

This chapter provides solutions for the most common issues you may encounter while using the RTL-SDR V3 Console. Issues are grouped by category: installation, bridge connection, audio, signals, and decoders.

Installation Issues

npm install fails with "ENOENT: no such file or directory"

You are running `npm install` in a directory without a `package.json` file. Make sure you have cloned the repository and `cd'd` into it before running `npm install`. If you are trying to install the bridge, download both `bridge.mjs` AND `package.json` from the Connection panel into the same folder.

npm run dev fails with "Cannot find module"

Dependencies are not installed. Run `npm install` first, then `npm run dev`. If the error persists, delete `node_modules` and `package-lock.json`, then run `npm install` again.

Page loads but shows a white screen

Check the browser console (F12 -> Console) for JavaScript errors. Common causes: browser cache serving old code (try `Ctrl+Shift+R` for hard refresh), JavaScript disabled, or a syntax error in custom modifications. The dev server log at `/home/z/my-project/dev.log` may also show errors.

Bridge Connection Issues

usb_open error -3 (Linux)

Your user does not have permission to access the USB device. Apply the `udev` rules from Chapter 5 Step 2, then log out and log back in. If that fails, use the `sudo` fallback: `sudo rtl_tcp -s 2400000`. The bridge itself does not need root - only `rtl_tcp` does.

Bridge shows "rtl_tcp connection closed, reconnecting"

The dongle was unplugged or another SDR app is holding it. Close any other SDR software (SDR#, GQRX, CubicSDR, etc.), then restart `rtl_tcp`. The bridge will automatically reconnect within 2 seconds.

Bridge shows "tuner: RTL-SDR (tuner 83886080)" instead of "R820T"

This was a bug in an earlier version of the bridge that read the `rtl_tcp` handshake with the wrong byte order. The number 83886080 is actually 5 in big-endian, which is the R820T tuner ID. Update to the latest bridge version (re-download `bridge.mjs` from the Connection panel).

Web app shows OFFLINE even with bridge running

Check the bridge URL in the Hardware Source panel. If the bridge is on a different PC, enter that PC's IP address (e.g. `ws://192.168.1.50:8080`). If you are accessing the web app via HTTPS (the cloud preview), you must use `wss://` and start the bridge with `--tls`. Browsers block `ws://` from HTTPS pages except to localhost.

SyntaxError: Invalid regular expression flags

The bridge.mjs file is corrupted - the leading # character of the shebang line (`#!/usr/bin/env node`) was stripped during copy-paste. Fix it with: `sed -i '1s/^!/#/' ~/sdr-bridge/bridge.mjs`. Then verify with: `head -1 ~/sdr-bridge/bridge.mjs` (should show `#!/usr/bin/env node`).

Audio Issues

No audio even with signal present

Check the audio activity meter in the transport bar. If it shows 0%, the demodulator is producing silence - verify you are on the correct demod mode (WFM for broadcast FM, AM for airband, etc.) and the signal is centered in the filter window (amber dashed lines on spectrum). If the meter shows activity but you hear nothing, check your browser's audio output and system volume.

Audio sounds clipped or choppy

The browser's WebSocket throughput tops out around 2-3 MB/s on a clean LAN. At 2.4 Msps, that is 4.8 MB/s of IQ data - too much for some connections. Try a lower sample rate: 1.024 Msps or 0.24 Msps. The bridge will report overruns in the connection panel if it is dropping samples.

Audio sounds distorted or off-pitch

Your dongle's PPM correction is off. Tune to a known exact frequency (e.g. a local FM station's licensed frequency from the FCC database) and adjust the PPM slider in the RF & Audio panel until the audio sounds natural. Once calibrated, the PPM value is constant for all frequencies.

Audio works in simulated mode but not real mode

The simulated audio engine and real audio engine use different code paths. In real mode, the audio engine needs to receive demodulated PCM frames from the real SDR source. If the activity meter shows 0%, the real audio path is not being fed. Verify you have clicked AUDIO ON after switching to real mode (the AudioContext needs a user gesture to start).

Signal Issues

Spectrum shows only noise, no signals

Check your antenna connection. The stock RTL-SDR whip should pick up at least some FM broadcast stations. If you see only noise: (1) verify the antenna is firmly connected to the MCX jack, (2) try a different antenna, (3) check that you are not in a shielded location (basement, metal building). Try tuning to a known strong FM station (e.g. 91.5 MHz, 101.5 MHz) - you should see a clear peak.

Strong signal but no decode

Each decoder requires specific conditions: RDS needs WFM mode and a station that broadcasts RDS; ADS-B needs 1090 MHz and an aircraft overhead; APT needs a satellite pass overhead. Check the decoder's chapter for specific requirements. Also verify the PPM correction - some decoders (RDS, HD Radio) are sensitive to frequency error.

Hydration error in browser console

If you see "Hydration failed because the server rendered text didn't match the client", this is usually caused by a browser extension (Grammarly, LastPass) injecting attributes into the HTML before React loads. The fix is to add `suppressHydrationWarning` to the body tag. This has been done in the latest version of the app - if you still see it, you may have a cached old version (try Ctrl+Shift+R).

High CPU usage

The real-SDR pipeline runs an FFT + demodulator + decoders on every IQ block. At 2.4 Msps, this consumes about 30-50% of one CPU core on a modern machine. If CPU usage is excessive: (1) reduce sample rate to 1.024 Msps, (2) close decoder panels you are not using (decoders auto-pause when their panel is not subscribed), (3) reduce the spectrum size if you have customized it.

Decoder-Specific Issues

RDS overlay shows "Waiting for RDS data" indefinitely

The signal is too weak or the station does not broadcast RDS. Try a stronger station (look for a tall peak on the spectrum). Some community and college stations do not carry RDS. Also check PPM correction - RDS is sensitive to frequency error.

ADS-B shows no aircraft

Verify you are tuned to 1090 MHz with sample rate 2.4 Msps. The antenna must be vertical and have a clear view of the sky. Try a different antenna position or upgrade to a 1090 MHz antenna. Check flightradar24.com to verify aircraft are in your area - if no planes are nearby, you will not decode any.

APT image is slanted or curved

Your PPM correction is off. Adjust the PPM slider in the RF & Audio panel until the image straightens out. Start at 0 PPM and adjust in steps of 5-10 PPM until the sync locks consistently. Once calibrated, the value is constant for all frequencies.

GPS shows no satellites

You need an active GPS antenna (with built-in LNA) - the stock whip will NOT work for GPS. Also ensure the antenna has a clear view of the sky - indoors you may get 0-2 satellites. Try moving the antenna to a window or outdoors.

Chapter 24: Glossary

This glossary defines the radio and SDR terms used throughout this documentation. Terms are listed alphabetically. If you encounter a term not defined here, Wikipedia and the ARRL Handbook are excellent additional references.

Term	Definition
ACARS	Aircraft Communications Addressing and Reporting System. Digital text messaging between aircraft and dispatchers on 131.55 MHz.
ADS-B	Automatic Dependent Surveillance-Broadcast. Aircraft transponder system broadcasting position/altitude on 1090 MHz.
AGC	Automatic Gain Control. Circuit that automatically adjusts receiver gain to maintain constant output level.
ALFN	Absolute Frame Number. HD Radio's GPS-locked time reference, counting milliseconds since GPS epoch (Jan 6, 1980).
AM	Amplitude Modulation. Encoding information in the amplitude of a carrier wave.
APT	Automatic Picture Transmission. NOAA weather satellite image format on 137 MHz.
BCH	Bose-Chaudhuri-Hocquenghem. A class of error-correcting codes used in POCSAG.
BER	Bit Error Rate. Fraction of received bits that are incorrect.
BPSK	Binary Phase Shift Keying. Digital modulation using two phase states (0 and 180 degrees).
Biquad	Second-order IIR filter used for low-pass, high-pass, band-pass, and notch filtering.
C/A code	Coarse/Acquisition code. The civil GPS spread-spectrum code at 1.023 Mchips/s.
CADU	Channel Access Data Unit. CCSDS frame structure used in satellite communications.
CDMA	Code Division Multiple Access. Multiple transmitters share the same frequency using unique spread-spectrum codes.
C/N0	Carrier-to-noise density ratio. GPS signal strength metric in dB-Hz.
CPR	Compact Position Reporting. ADS-B's encoding scheme for lat/lon in 35 bits each.
CRC	Cyclic Redundancy Check. Error-detecting code appended to data frames.
CT	Clock Time. RDS field carrying UTC time and date.
CW	Continuous Wave. Morse code transmission (an unmodulated carrier keyed on and off).
dBFS	Decibels relative to Full Scale. Signal level relative to the maximum the SDR can represent.
DBPSK	Differential BPSK. BPSK where the phase difference between consecutive symbols carries the data.
Doppler shift	Frequency change due to relative motion between transmitter and receiver.
EVM	Error Vector Magnitude. Measure of modulation quality (lower is better).
FEC	Forward Error Correction. Redundant data added to allow error correction at the receiver.

Term	Definition
FM	Frequency Modulation. Encoding information in the frequency of a carrier wave.
FSK	Frequency Shift Keying. Digital modulation using discrete frequency changes.
GMDSS	Global Maritime Distress and Safety System. International maritime emergency communication system.
GOES	Geostationary Operational Environmental Satellite. US weather satellite system.
GPS	Global Positioning System. US satellite navigation system on 1575.42 MHz.
HD Radio	Hybrid Digital Radio. North American digital broadcasting standard (NRSC-5).
HDLC	High-Level Data Link Control. Frame protocol used in Inmarsat STD-C and many other protocols.
HDC	High-Definition Coding. Proprietary audio codec used in HD Radio (no open-source decoder).
ICAO	International Civil Aviation Organization. 24-bit unique aircraft address in ADS-B.
IIR	Infinite Impulse Response. A class of digital filters with feedback (e.g. biquad).
IQ samples	Complex baseband samples (In-phase + Quadrature) representing the received signal.
LFSR	Linear Feedback Shift Register. Used to generate GPS C/A codes.
LNA	Low-Noise Amplifier. Preamplifier placed at the antenna feed to overcome coax loss.
LSB	Lower Sideband. SSB mode keeping the lower sideband (frequencies below the carrier).
MSK	Minimum Shift Keying. FSK with deviation exactly half the bit rate (used in ACARS).
NCS	Network Coordination Station. Inmarsat's master station for each satellite region.
NFM	Narrow FM. FM with 5 kHz deviation, used for VHF/UHF voice.
NRSC-5	Standard governing HD Radio in North America.
OFDM	Orthogonal Frequency Division Multiplexing. Multi-carrier modulation used in HD Radio and WiFi.
OOOI	Out/Off/On/In. ACARS flight phase reports.
PCM	Pulse Code Modulation. Digital audio encoding (used in WAV files).
PI code	Program Identification. 4-hex RDS code uniquely identifying a station.
POCSAG	Post Office Code Standardization Advisory Group. Pager protocol on 929-932 MHz.
PPM	Parts Per Million. Frequency error unit (1 PPM at 100 MHz = 100 Hz error).
PRN	Pseudo-Random Noise. GPS satellite identifier (PRN-1 through PRN-32).
PS	Program Service. RDS field carrying the 8-character station name.
PSK	Phase Shift Keying. Digital modulation using discrete phase changes.
PTY	Program Type. RDS field (0-31) indicating station format (News, Rock, etc.).

Term	Definition
Q factor	Quality factor of a filter. Higher Q = narrower bandwidth.
QAM	Quadrature Amplitude Modulation. Combined amplitude + phase modulation.
QFH	Quadrifilar Helix. Antenna with circular polarization for satellite reception.
QPSK	Quadrature PSK. PSK with 4 phase states (2 bits per symbol).
RDS	Radio Data System. 1187.5 bps data subcarrier on FM broadcast (57 kHz).
RHCP	Right-Hand Circular Polarization. Antenna polarization used by GPS and weather satellites.
RT	Radio Text. RDS field carrying up to 64 characters of free text.
S-meter	Signal strength meter (S0 through S9+60 dB).
SNR	Signal-to-Noise Ratio. Ratio of signal power to noise power.
SSB	Single Sideband. AM with one sideband removed for efficiency (USB or LSB).
STD-C	Standard-C. Inmarsat's low-bit-rate messaging service on L-band.
Squelch	Circuit that mutes audio when signal drops below a threshold.
TA	Traffic Announcement. RDS flag indicating traffic news.
USB	Upper Sideband. SSB mode keeping the upper sideband (frequencies above the carrier).
Viterbi	Convolutional code decoding algorithm used in many digital communications.
WAV	Waveform Audio File Format. Standard uncompressed audio file format.
WFM	Wide FM. FM with 75 kHz deviation, used for broadcast.

Chapter 25: Limitations & What's Not Possible

This chapter provides an honest assessment of what the RTL-SDR V3 Console can and cannot do. Some limitations are inherent to the RTL-SDR hardware; others are implementation choices in the console. Understanding these limits helps you decide when to use this tool and when to upgrade to more capable hardware.

Hardware Limitations

Frequency Range

The RTL-SDR V3 covers 24 kHz to 1.75 GHz. This is a wide range but excludes several interesting bands:

Band	Frequency	What You're Missing
S-band	2.3-2.4 GHz	SiriusXM satellite radio
C-band	4-8 GHz	Some satellite downlinks
X-band	8-12 GHz	Military and satellite
Ku-band	12-18 GHz	DirecTV, Dish Network
Ka-band	26-40 GHz	Starlink user terminals

To receive these higher bands, you need an SDR with a higher frequency range. The HackRF One covers up to 6 GHz (covers SiriusXM), the LimeSDR covers up to 3.8 GHz, and the PlutoSDR covers up to 6 GHz. None of these can reach Starlink's user terminal frequency (12 GHz) - that requires specialized microwave SDRs.

Sample Rate

The RTL-SDR V3's maximum reliable sample rate is 3.2 Msps (theoretical max 3.2 Msps). This limits the maximum signal bandwidth you can receive to about 3.2 MHz. Wide signals like DVB-T (6-8 MHz), WiFi (20+ MHz), and LTE (up to 20 MHz) cannot be captured in full. Higher sample rates are possible with the RTL-SDR but become increasingly lossy due to USB 2.0 bandwidth limits.

Bit Depth

The RTL-SDR V3 produces 8-bit IQ samples. This gives about 48 dB of dynamic range, which is sufficient for most signals but limits the ability to receive weak signals in the presence of strong ones. For comparison, the Airspy R2 has 12-bit ADCs (72 dB dynamic range), and the SDRplay RSPdx has 14-bit ADCs (84 dB dynamic range). The difference matters when you are trying to hear a weak signal adjacent to a strong one.

Software Limitations

HD Radio Audio

The console decodes HD Radio SIS (station information) but not HD Radio audio. HD Radio audio uses the HDC (High-Definition Coding) codec, which is proprietary and has no open-source decoder. Even the open-source nrsc5 project (the leading HD Radio decoder) had to reverse-engineer parts of the codec. Implementing HDC in JavaScript would require porting thousands of lines of C code and may face legal challenges.

GPS Position Fix

The console tracks GPS satellites and decodes nav bits, but does NOT compute a position fix. A full position fix requires parsing ephemeris data (satellite positions) from subframes 2 and 3 of the nav message, then performing trilateration using the pseudoranges from at least 4 satellites. This is mathematically straightforward but requires careful implementation of the trilateration algorithm. It is left as a future enhancement.

Meteor M2 Image Decompression

The console decodes Meteor M2 LRPT signal and assembles CCSDS CADU frames, but does NOT decompress the image data. Decompression requires JPEG-LS, a specialized lossless compression standard. Implementing JPEG-LS in JavaScript is possible but would require porting a C library. Instead, the console lets you download the raw CADU bytes for offline decompression with SatDump or similar tools.

DAB/DAB+ (European Digital Radio)

The console does not decode DAB/DAB+, the European digital radio standard. DAB uses OFDM on 174-240 MHz with a completely different protocol than HD Radio. Adding DAB support would require implementing a new OFDM demodulator, MP2/AAC audio decoder, and DAB-specific protocol stack. This is feasible but significant work.

Trunked Radio

The console does not track trunked radio systems (Motorola, P25 Phase II, EDACS, etc.). Trunking requires following talkgroup assignments across multiple frequencies, which is a complex state machine. For trunked radio monitoring, use dedicated software like SDRTrunk, Unitrunker, or DSD+.

What's Impossible (Not Just Hard)

SiriusXM Satellite Radio

SiriusXM broadcasts on 2.3 GHz (S-band), well above the RTL-SDR V3's 1.75 GHz cutoff. Even with a HackRF (6 GHz), SiriusXM uses the PAC (Perceptual Audio Coder) codec which is fully proprietary - there is no open-source decoder anywhere. SiriusXM also uses conditional access (encryption keyed to your receiver's ESN). Receiving SiriusXM with any SDR is impossible without the decryption keys, which only SiriusXM-controlled hardware has.

Starlink

Starlink user terminals receive on 10.7-12.7 GHz (Ku-band), far above the RTL-SDR's range. Even specialized microwave SDRs that can receive Ku-band would face Starlink's proprietary protocols, beam-hopping, and encryption. There is no way to decode Starlink traffic with any open-source tool.

Encrypted Communications

Many digital communications are encrypted: P25 with encryption (AES-256), TETRA with encryption, DMR with privacy keys, GSM/UMTS/LTE cellular, encrypted WiFi. The console does not and will not decrypt any encrypted communication. In most countries, decrypting communications you are not authorized to receive is illegal.

When to Upgrade

If you find yourself hitting the RTL-SDR V3's limits, consider upgrading to:

SDR	Frequency Range	Sample Rate	Bits	Best For	Cost
RTL-SDR V3	24 kHz - 1.75 GHz	3.2 Msps	8	Beginner, general listening	\$30
Airspy R2	24 MHz - 1.8 GHz	10 Msps	12	Weak signal, HF	\$170
SDRplay RSPdx	1 kHz - 2 GHz	10 Msps	14	VLF/LF, multi-antenna	\$200
HackRF One	1 MHz - 6 GHz	20 Msps	8	Wide band, transmit	\$300
PlutoSDR	70 MHz - 6 GHz	56 Msps	12	5G, WiFi, transmit	\$200
LimeSDR	100 kHz - 3.8 GHz	61 Msps	12	MIMO, transmit	\$300

The RTL-SDR V3 is the best value for getting started with SDR. For most users, it covers the interesting bands (FM, airband, ADS-B, weather satellites, pagers, GPS) at a fraction of the cost of higher-end SDRs. Upgrade when you have a specific need that the RTL-SDR cannot meet.

Appendix A: Bridge Protocol Reference

This appendix provides a complete reference for the bridge's wire protocol, useful if you want to write your own client or extend the bridge.

Command-Line Flags

Flag	Default	Description
--rtl-host	127.0.0.1	rtl_tcp host
--rtl-port	1234	rtl_tcp port
--ws-host	0.0.0.0	WebSocket bind address
--ws-port	8080	WebSocket port
--tls	(off)	Enable wss:// (auto-generates self-signed cert)
--cert	(auto)	TLS certificate path (with --tls)
--key	(auto)	TLS private key path (with --tls)
--http-port	8081	HTTP server port (recordings + presets)

WebSocket Commands (Client to Server)

All commands are JSON text frames:

```
{"type":"set_frequency","hz":91500000}  
Tune to a frequency in Hz.  
  
{"type":"set_sample_rate","hz":2400000}  
Set the sample rate in Hz.  
  
{"type":"set_gain","db":30}  
Set manual gain to 30 dB.  
  
{"type":"set_gain","db":"auto"}  
Enable automatic gain control.  
  
{"type":"set_ppm","ppm":15}  
Set frequency correction to +15 PPM.  
  
{"type":"start"}  
Start streaming IQ data.  
  
{"type":"stop"}  
Stop streaming IQ data.  
  
{"type":"status"}  
Request an immediate status update.  
  
{"type":"start_recording"}  
Start recording IQ to disk on the bridge.  
  
{"type":"stop_recording"}
```

```
Stop recording and finalize the file.
```

WebSocket Messages (Server to Client)

Status Message

Sent every 500 ms and on demand:

```
{
  "type": "status",
  "payload": {
    "connected": true,
    "deviceName": "RTL-SDR V3 (R820T2)",
    "frequency": 91500000,
    "sampleRate": 2400000,
    "gainDb": 30,
    "ppm": 0,
    "gains": [],
    "overruns": 0,
    "uptime": 42.3,
    "recording": null
  }
}
```

Recording Complete Message

Sent when recording stops:

```
{
  "type": "recording_complete",
  "payload": {
    "path": "/path/to/recording.raw",
    "bytes": 52428800,
    "duration": 10.9,
    "sampleRate": 2400000,
    "frequency": 91500000
  }
}
```

Binary IQ Frame Format

Each binary frame has a 16-byte little-endian header followed by raw IQ bytes:

Offset	Size	Type	Description
0	4	uint32 LE	Sample rate in Hz
4	4	uint32 LE	Frequency - low 32 bits
8	4	uint32 LE	Frequency - high 32 bits (always 0 for RTL-SDR)
12	4	uint32 LE	Timestamp (ms, truncated to 32 bits)
16+	variable	bytes	Interleaved unsigned 8-bit I/Q pairs

Each IQ sample is two bytes: I (in-phase) and Q (quadrature), both unsigned 8-bit centered on 127. To convert to signed float: subtract 128, divide by 128. Frames are sent in 32 KB chunks (16K

complex samples) every 50 ms.

HTTP Endpoints

Endpoint	Method	Description
/health	GET	Returns JSON {ok:true, ws:'ws' \wss', port:8080}
/recordings	GET	Returns JSON array of saved recordings
/recordings/{filename}	GET	Downloads a recording file
/presets	GET	Returns saved bookmarks JSON (or empty)
/presets	PUT	Saves bookmarks JSON to presets.json

RTL-SDR Command Bytes

The bridge forwards commands to rtl_tcp using its native binary protocol:

Byte	Constant	Value Format	Description
0x01	SET_FREQUENCY	uint32 BE (Hz)	Tune to frequency
0x02	SET_SAMPLE_RATE	uint32 BE (Hz)	Set sample rate
0x03	SET_GAIN_MODE	uint32 BE (0/1)	0=manual, 1=auto
0x04	SET_GAIN	uint32 BE (dB*10)	Manual gain (300 = 30.0 dB)
0x05	SET_FREQ_CORRECTION	uint32 BE (PPM)	Frequency correction

Each command is 5 bytes: 1 byte command code + 4 bytes big-endian value. The bridge handles the conversion from JSON commands to rtl_tcp binary commands automatically.

Appendix B: Frequency Reference

This appendix provides a quick reference for all frequencies the console can receive, organized by band. Use it to plan your antenna setup and know what to expect at each frequency.

Broadcast Bands

Band	Frequency	Demo d	Decoder	Antenna
FM Broadcast	87.5-108 MHz	WFM	RDS, HD Radio	Stock whip
AM Broadcast (MW)	530-1710 kHz	AM	-	Long wire
Shortwave Broadcast	2-30 MHz	AM	-	Long wire

Aviation Bands

Band	Frequency	Demo d	Decoder	Antenna
Airband Voice	118-137 MHz	AM	ACARS (131.55)	Vertical dipole
ADS-B	1090 MHz	RAW	ADS-B	1090 MHz antenna
ACARS	131.55 MHz	AM	ACARS	Stock whip

Marine Bands

Band	Frequency	Demo d	Decoder	Antenna
Marine VHF	156-162 MHz	NFM	-	Vertical whip
AIS	161.975/162.025 MHz	NFM	-	Vertical whip
Ch 16 Distress	156.8 MHz	NFM	-	Vertical whip

Weather

Band	Frequency	Demo d	Decoder	Antenna
NOAA WX Radio	162.4-162.55 MHz	NFM	-	Stock whip
NOAA APT	137.1/137.5/137.9 MHz	WFM	APT	V-dipole or QFH
Meteor M2	137.1/137.9 MHz	WFM	Meteor LRPT	V-dipole or QFH
GOES HRIT	1685.7/1694.1 MHz	RAW	GOES HRIT	10-turn helical + LNA

Ham Radio Bands

Band	Frequency	Demod	Notes
160m	1.8-2.0 MHz	LSB/CW	Night, long wire
80m	3.5-4.0 MHz	LSB/CW	Night, long wire
40m	7.0-7.3 MHz	LSB/CW	Day/night, dipole
20m	14.0-14.35 MHz	USB/CW	Day, dipole
15m	21.0-21.45 MHz	USB/CW	Day, dipole
10m	28.0-29.7 MHz	USB/CW	Day, dipole
2m	144-148 MHz	NFM	Stock whip
70cm	420-450 MHz	NFM	Stock whip

Pager Bands

Band	Frequency	Demo d	Decoder	Antenna
US Pager	929-932 MHz	NFM	POCSAG	900 MHz Yagi
VHF Pager	138-174 MHz	NFM	POCSAG	Stock whip

Satellite Bands

Band	Frequency	Demo d	Decoder	Antenna
Inmarsat STD-C	1537-1545 MHz	RAW	STD-C	Helical + LNA
GPS L1 C/A	1575.42 MHz	RAW	GPS	Active GPS antenna
GOES HRIT	1685.7 MHz	RAW	GOES	10-turn helical + LNA

Other Useful Frequencies

Frequency	Description	Demod
10 MHz	WWV time signal (Fort Collins, CO)	AM
15 MHz	WWV time signal	AM
60 kHz	WWVB time signal (NIST)	CW
Marine Ch 16	156.8 MHz	NFM
CB Channel 19	27.185 MHz	AM

This reference covers the most commonly received frequencies. For a complete list of frequencies in your area, consult the FCC database (US) or your local regulator's database. Online resources like radioreference.com also provide comprehensive frequency lists with user-submitted notes.